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# Intra-annual compositions and diversity variations of waterbird communities in China

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ABSTRACT

Waterbirds are often used as indicators of wetland biodiversity and ecosystem health due to their sensitivity to environmental changes. However, most studies have focused on long-term variations of waterbird communities, while the intra-annual changes have rarely been investigated. Here, we analyzed the seasonal dynamics of compositions and the intra-annual variations of species richness and abundance of waterbird communities among the nine major river basins of China, and focused on the effects of different residential types and taxonomic groups on the diversity variations. In most of the nine basins, the waterbird communities in spring and autumn showed high similarity, while those in summer and winter were quite different. Such a seasonal cycle of species composition was caused by the seasonal migration of waterbirds. The season with higher species richness shifted from summer to winter as the latitude of the basin decreased within China, but almost all basins had the lowest community abundance in summer. This was because the variation of richness was directly influenced by the variation in species composition, but the variation of abundance was more influenced by the population size of different waterbird assemblages. Our results emphasized the importance of seasonal dynamics of waterbird communities to the understanding of their migration patterns, and highlighted the impact of different waterbird assemblages on intra-annual community variations. Additionally, we also noted the potential effects of population size and behaviors at different parts of the waterbird life circle on the observations of them. The information gathered in this study will provide guidance for the conservation of waterbird species and their habitats.

## 1. Introduction

Birds play a vital role in ecosystem dynamics and are highly sensitive to environmental changes (Li et al., 2021; Sekercioglu, 2006;

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Whelan et al., 2008). Compared to other bird species, waterbirds are especially responsive to environmental variations (Pavón-Jordán, 2017), and are frequently used as indicators in biological and ecosystem monitoring programs (Bryce et al., 2002; Gregory and van Strien, 2010), because of their vital roles in indicating wetland biodiversity and ecosystem characteristics (Amano et al., 2018; Tallei et al., 2021), habitat quality (Zou et al., 2019), and wetland ecosystem conservation (Ghimire et al., 2021; Siddig et al., 2016). It is found that global waterbird populations have continued to decline (Amano et al., 2018; Li et al., 2021) due to the habitat destruction and climate change (Mantyka-Pringle et al., 2015; Opdam and Wascher, 2004). However, the trends of the waterbird community under different research scales and environmental conditions were inconsistent, such as the abundance of waterbird populations decreased at lower latitudes but increased at higher latitudes (Amano et al., 2020), and for local scales, the waterbird community increased in abundance and diversity because of the protection of wetlands (Li et al., 2021). Therefore, Understanding the spatiotemporal variations of waterbird communities at the appropriate scale is of great significance for the conservation of waterbird species.

The composition, abundance and distribution of bird assemblages are directly associated with the fluctuations in habitat, climate and resource availability (Gaüzère et al., 2015; Lindström et al., 2013; Tallei et al., 2021). Although the variations in the temporal dimension of waterbird communities have not gone unnoticed, the changes in temporal periods of them are rarely addressed (Seoane et al., 2013). The majority of studies have shown the long-term decline of waterbird community diversity and abundance with declining wetland quality (Jia et al., 2018; Li et al., 2022; Zhou et al., 2020), while wetland restoration leads to community enhancement (Fan et al., 2021; Rajpar et al., 2022; Song et al., 2024). However, these studies often rely on environmental factors that are assumed to have constant effects throughout the year, when in fact whose intra-annual or seasonal variation can also lead to variations in the waterbird community within the year (Seoane et al., 2013). The migratory birds move across different regions throughout their lifetime to acquire seasonally available food resources and avoid stressful conditions (Jahn et al., 2020), resulting in the community variation with seasonal regularities (Butchart et al., 2008). During the breeding season, birds spatially restrict their mating to a central place in relation to vegetation structure, while they explore a greater variety of habitats over larger areas in winter (Lopez de Casenave et al., 2008; Seoane et al., 2013). However, most of the studies only focused on parts of the life circle (e.g., breeding, overwintering, or stopover sites) of waterbirds, rather than exploring the community variation throughout the year, which are important for understanding their migration patterns. Moreover, environmental factors affect waterbird communities over large time scales (Jia et al., 2018; Ma et al., 2010), but in shaping their intra-annual changes, especially of abundance, the population size of different taxonomic groups in communities also plays an important role. Therefore, there is an urgent need to analyze the intra-annual diversity and composition variations of waterbird communities, which is essential for understanding waterbird migration routes, predicting waterbird community responses to global climate change (Li et al., 2021), and developing effective policies for waterbird species conservation and habitat management (Ghimire et al., 2021; Li et al., 2021).

Given that geographic and climatic gradients help shape community composition at larger scales (Lindström et al., 2013), China has supported a large number of waterbird communities due to its vast territory, diverse climate, and wetland types (Wang et al., 2018). Coupled with the passage of three of the eight major global migratory flyways (East Africa-West Asia Migratory Flyway, Central Asia Migratory Flyway, and East Asia-Australasia Migratory Flyway), China occupies a pivotal position in global waterbird research and conservation (Wang et al., 2018). The regional differences of intra-annual variations of waterbird communities in China are foreseeable, because the diversity variations can be explained by the characteristics of the regional and seasonal environment (Seoane et al., 2013), and China has broad environmental gradient and distinctive seasons (Li et al., 2021; Liang et al., 2024). However, most existing studies on Chinese waterbirds have been limited to local scales, such as a single wetland or river (C. Wang et al., 2021,2021; Zou et al., 2019), but rarely conducted at the cross-basin scales. According to one of the established divisions of geographical regions of China, China can be divided into nine basins based on its major rivers. Due to factors such as location, topographical conditions, and climates, each basin has its unique natural environment and human activity patterns, which have resulted in wetland habitats for waterbirds with different types and areas. The waterbird communities vary with different habitat characteristics, but with important seasonal dynamics (González-Gajardo et al., 2009). Therefore, in this study, we selected the mainland and the two nearest and largest southern islands of China as the research region, according to the data from the Resource and Environmental Science and Data Center of the Chinese Academy of Sciences, to analyze the intra-annual variations of composition and diversity of regional waterbird communities in each of the nine major river basins. We hypothesized that: 1. The seasonal variation regularity of the waterbird community is primarily driven by species turnover caused by waterbird migration. 2. The intra-annual variations of species richness and abundance are led by different waterbird assemblages of communities among the nine basins. Moreover, in order to explain the differences in the composition and diversity of waterbird communities among different basins, we selected the wetland area and the ecological environment quality as environmental factors to briefly analyze their effects on regional waterbird communities. The waterbirds depend crucially on wetland habitats (Tavares et al., 2015), whose area is closely linked to resource abundance and distribution (Donnelly et al., 2020). The Ecological Environment Quality Index (EEQI) is a comprehensive environmental evaluation index that evaluates regional environmental quality from three levels of Pressure-State-Response. The EEQI value of nine basins was calculated by the EEQI value of each province in China (Ren and Lv, 2019).

## 2. Methods

### 2.1. Study area

China has a vast territory, complex terrain, and diverse climate conditions (An et al., 2022; Chen et al., 2021). The uneven distribution of rivers, different wetland types, and seasonal climate variations have provided various aquatic habitats to waterbirds (Xu et al., 2014; Zhao and He, 2001). According to the spatial distribution of major rivers in China, the Chinese nine major river basins

include the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB) (Fig. A.1, Table A.1). The boundary data for each basin was obtained from the Resource and Environmental Science and Data Center of the Chinese Academy of Sciences (http://www.resdc.cn).

### 2.2. Waterbird data

This study utilized the Chinese nationwide bird-watching reports downloaded from China Bird Report (http://www.birdreport.cn/ ). There were totally 12048 reports uploaded by bird-watchers from 2005 to June 2020, with 1358 bird species observed. Each report contained the time, location, species, and individual numbers. After eliminating years with insufficient data, the dataset of all reports from May 2014 to June 2020 was determined for data analysis, with a total of 253 waterbird species observed, according to the list of waterbirds in China (Liu et al., 2019)(Fig. A.1, Table A.2). According to the information on the migrant status and biogeographic distribution of Chinese waterbirds obtained from a dataset on the life-history and ecological traits of Chinese birds (Y. Wang et al., 2021), we calculated the residential type of each species in each basin. In Wang's dataset, the residential type of each waterbird species was categorized into passage migrant (P), summer visitor (S), winter visitor (W), resident (R), or vagrant visitor (V) according to their migrant status in each part of the zoogeographical divisions of China. To map the residential types to each of Chinese nine major river basins, all parts of the zoogeographical divisions intersected with the basin were considered to calculate the residential type of waterbird species. If the species has more than one migrant status in one basin, its residential type will be proportionally divided with coefficients sum to 1 when calculating the residential types of the waterbird community.

### 2.3. Data analysis

The observation data was divided into nine datasets according to the boundary data of the nine major river basins in China. We merged the data for the six years by month to conduct seasonal (spring (Mar-May); summer (Jun-Aug); autumn (Sep-Nov); winter (Dec-Feb)) and regional analysis of Chinese waterbird communities. In each month for each basin, we added up the waterbird species and their abundance of the six years (calculated over the period 2014–2020), and took the sum of observation times into consideration to mitigate the impact of discrepancies in data quantity across the nine basins to data analysis.

We selected the dominant assemblage of each basin to assess the differences of composition in waterbird communities among the nine basins, considering to avoid the influence of occasional species. The dominant assemblage can characterize the species composition of the community, thus is suitable for comparing the composition of different communities. We identified dominant assemblages using the Mcnaughton dominance index (McNaughton, 1967), whose formula was applied:

$$Y_i = \frac{n_i}{N} f_i$$

where $n_i$ is the total abundance of waterbird species $i$ in all basins, $N$ is the total abundance of all waterbird species in all basins, and $f_i$ is the occurrence frequency of the species $i$ across all basins. A waterbird species with $Y_i \geq 0.02$ is identified as a dominant species.

The species composition similarities of dominant assemblages between pairs of basins were calculated using the Jaccard index (Jaccard, 1912). The index was calculated as follows:

$$J_i = \frac{C_i}{D_i + E_i - C_i}$$

where $C_i$ is the number of waterbird species identified as dominant in both regions, $D_i$ is the number of waterbird species identified as dominant in region A, and $E_i$ is the number of waterbird species identified as dominant in region B.

The non-metric multidimensional scaling analysis (NMDS) was used to visualize the differences in waterbird community composition among seasons. The abundance used in NMDS was expressed as the mean abundance of one observation of each month in each year.

In this study, we chose species richness and abundance to demonstrate the intra-annual variation of waterbird communities. Prior to analysis, the abundance of each waterbird community was log10 transformed to constrain the impact of extremely high values. We used the generalized additive models (GAMs, (Trevor and Robert, 1986)) to perform the regression analysis between observation months and the species richness or abundance of waterbird communities in each basin. The model was built in which the response variable was the richness or abundance, regressed month expressed as a factor (1–12 corresponding to Jan-Dec). We used the sine and cosine functions to constrain the time of year as a circular factor considering the circulation of months, and took the number of observations as a covariate to handle the impact of uneven volume of observation data. For the illustration of intra-annual variation, we chose 50 times as the observation frequency.

To compare the effects of different residential types and orders of waterbirds on the overall intra-annual diversity variations of waterbird communities, we calculated the intra-annual variation of each assemblage also by using GAMs. We believe that the assemblages with higher index values of richness or abundance, and more similar variation curves to that of the community, have a greater impact on the community. To support this assessment method, we also used the Simper analysis (CLARKE, 1993) to perform pairwise comparisons of communities in two adjacent months in each basin, and displayed the assemblage with the most contribution to the differences in these two communities, which was assessed using the "average" value.

To evaluate the effect of regional differences of wetland area and EEQI on regional differences of waterbird communities among different river basins, we calculated the Euclidean distances matrix for environmental factors and the Jaccard distances matrix for dominant assemblages of regional waterbird communities, and used the generalized least squares (GLS, (Carroll and Ruppert, 1988)) model with the correlation structure between pairs of communities defined by mixed linear predictor equation (MLPE, (Clarke et al., 2002)) correlation. We also used the Local Regression to fit smooth curves of environmental factors and diversity of waterbird communities among nine basins, to visualize the trend of community diversity with environmental indices.

All the statistical analyses and diagrams were conducted using R version 4.3.1 (R Core Team, 2023).

## 3. Results

### 3.1. Differences of regional compositions and seasonal dynamics of waterbird communities

The 253 waterbird species selected for this study belonged to 11 orders (Table A.2). The four richest orders (Charadriiformes, Anseriformes, Pelecaniformes, Gruiformes) accounted for 85.38 % of all species, the Charadriiformes and Anseriformes are the most species-rich orders and are the main components of waterbird communities throughout the country, accounting for 43.08 % and 19.76 % of the total species, respectively. Of all observed waterbird species, 65 species (25.69 %) were registered as globally threatened according to the IUCN Red List (IUCN, 2023). At the basin scale, a total of 146 species were recorded in the ConB, 153 species in the SLRB, 158 species in the HaiRB, 147 species in the YelRB, 153 species in the HuaiRB, 193 species in the YzRB, 108 species in the SwB, 177 species in the SeB, and 173 species in the PeaRB. In total, 129 species with the value of Mcnaughton dominance index higher than 0.02 were classified as dominant species in at least one of the Chinese nine major river basins (Table A.3), among which only 12 species including *Mareca penelope*, *Anas platyrhynchos*, *Anas crecca*, *Aythya fuligula*, *Tachybaptus ruficollis*, *Podiceps cristatus*, *Ardea cinerea*, *Ardea alba*, *Phalacrocorax carbo*, *Fulica atra*, *Charadrius dubius* and *Chroicocephalus ridibundus* were recorded in all basins.




**Fig. 1.** The value of Jaccard similarity index representing the similarity degree between the species compositions of waterbird dominant assemblages in two river basins. The index ranges from 0 to 1, where a value of 1 indicates two completely identical assemblages, and a value of 0 indicates that the two assemblages are entirely dissimilar. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

The Jaccard similarity index was measured to represent differences in dominant species assemblages between nine basins. The highest similarity index was found between the YelRB and YzRB (0.71), followed by that between the ConB and YelRB (0.7). In contrast, the SLRB and SwB had the lowest value (0.22). In general, the similarity index values between each of the three basins (the SwB, SeB and PeaRB) and other basins were obviously lower compared to the values of other set-pairs (Fig. 1).

The results of the GLS model indicated a positive relationship but low correlation between the regional differences of wetland size (Standardized Coefficients = 0.117, t = 0.707, p = 0.485), EEQI (Standardized Coefficients = 0.250, t = 1.736, p = 0.092) and the regional differences of dominant species assemblages in waterbird communities among basin. Compared to that of wetland size, the regional differences of EEQI had a relatively higher correlation with composition differences of regional waterbird communities.

The ordination plots of most basins, excepting for the SwB, showed a clear separation or slight overlap between summer and winter waterbird communities, while there was a high overlap between spring and autumn waterbird communities (Fig. 2). The distances between the centroids of spring and autumn communities were clearly closer than that of summer and winter communities (Fig. A.2). However, the waterbird communities of all seasons highly overlapped with each other in the SwB.

### 3.2. *Intra-annual variations of species richness and abundance in waterbird communities*

The intra-annual species richness variations of waterbird communities varied across the nine major river basins of China, with the maxima of variation curves shifting from summer to winter as the latitudes of basins decreased (Fig. 3). Species richness of waterbird communities in the ConB, SLRB, HaiRB and YelRB exhibited intra-annual variations that could be described as unimodal curves, with

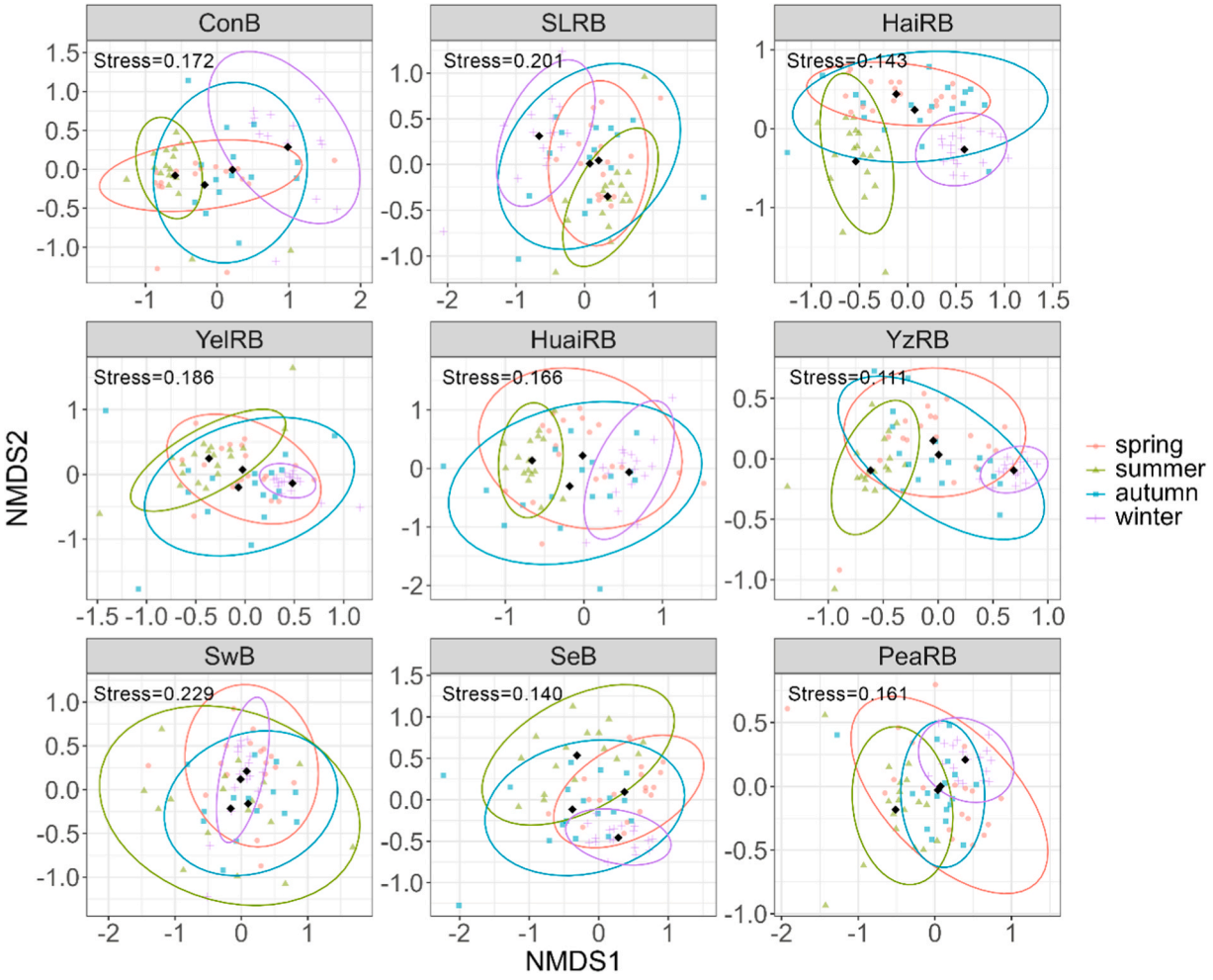



**Fig. 2.** Ordination plots of non-metric multidimensional scaling (NMDS) analysis in seasonal waterbird communities among the nine major river basins of China. Each point represents a waterbird community for one month in one year, with points that are closer together indicating more similar communities. The ellipses represent the range of seasonal waterbird communities. The black points are centroids of seasonal communities. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

the highest values occurring in summer. The variation curves of the HuaiRB, YzRB, SwB and PeaRB were U-shape with the lowest value occurring in summer. Species richness changed gently in the SeB month by month.

Regarding the intra-annual abundance variations of waterbird communities among the nine basins, besides the SLRB, all the other basins showed U-shaped curves with the lowest value occurring in summer (Fig. 4).

When comparing the intra-annual variation of species richness and abundance in the same basin, opposite trends were observed in the ConB, HaiRB and YelRB, and the SLRB also exhibited intra-annual trends of species richness and abundance which were less similar (Fig. 3; Fig. 4). In general, the basins located in southern China (SeB, SwB, PeaRB) exhibited lower overall values of species richness and abundance than other basins. The SwB had the lowest overall values for both richness and abundance compared to the other basins.

### 3.3. *Effects of different waterbird assemblages on intra-annual variations of communities*

The assemblages which affect the intra-annual variations of waterbird communities the most would have similar variation curves as that of the communities. Our results showed that the effect of each residential type on intra-annual variations of species richness in waterbird communities varied among basins (Fig. 5A). The variation curves of passage migrants and summer visitors, with relatively high values of the richness index, were similar to that of communities in the ConB, SLRB and YelRB. The residential type with the highest index values and the most similar variation curves to that of waterbird communities in the HaiRB and HuaiRB was the passage migrants, and in the SeB and PeaRB was the winter visitors. In the YzRB, the obvious seasonal variations of passage migrants, summer visitors and winter visitors shaped the intra-annual variation of community together, while there was no significant difference in the effect of passage migrants, summer visitors, winter visitors or residents on community variation curve in the SwB. Moreover, the effect of residents on intra-annual variations of communities was noticeably greater in the YzRB, SwB, SeB and PeaRB with relatively low latitudes than that in the other five basins.

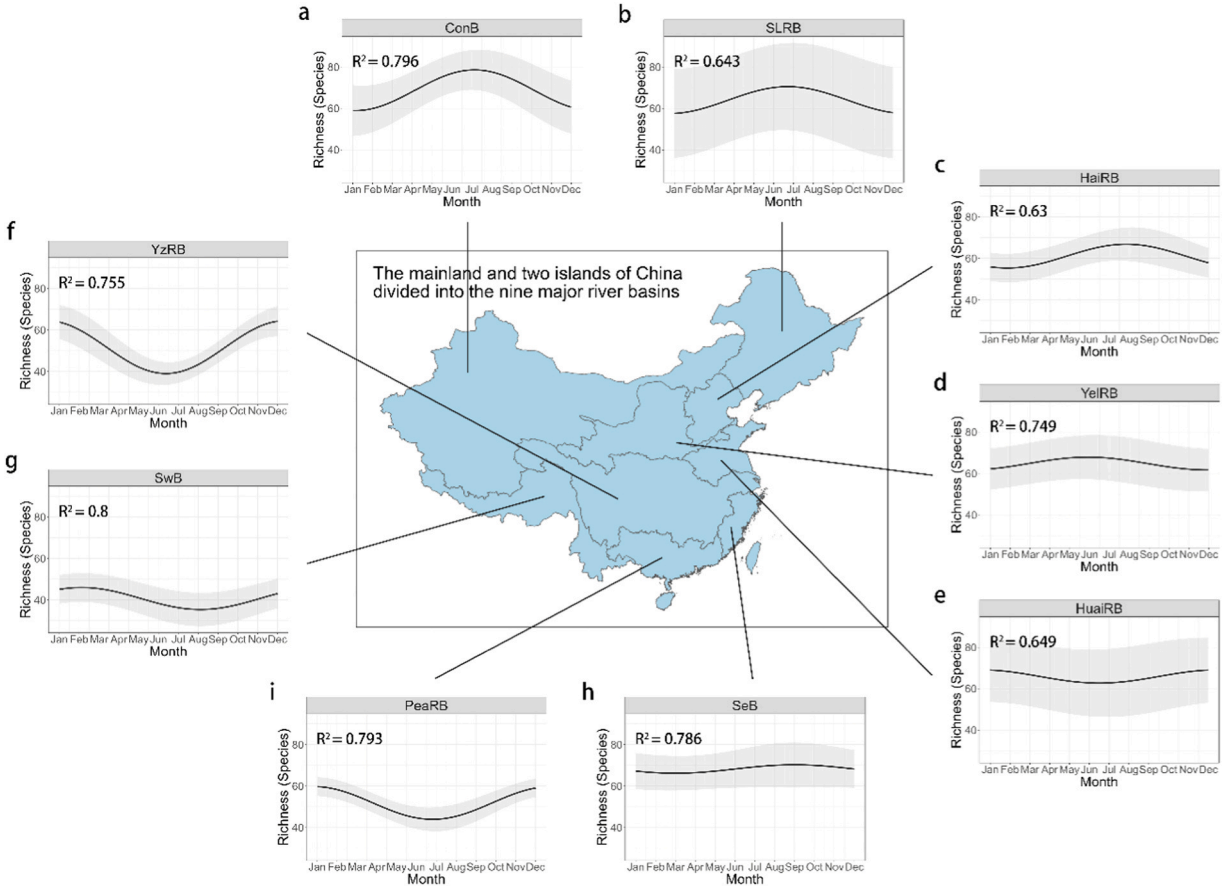



**Fig. 3.** Intra-annual variations of species richness in the waterbird community for each basin. a: the richness variation of the Continental Basin (ConB); b: the richness variation of the Songhua and Liaohe River Basin (SLRB); c: the richness variation of the Haihe River Basin (HaiRB); d: the richness variation of the Yellow River Basin (YelRB); e: the richness variation of the Huaihe River Basin (HuaiRB); f: the richness variation of the Yangtze River Basin (YzRB); g: the richness variation of the Southwest Basin (SwB); h: the richness variation of the Southeast Basin (SeB); i: the richness variation of the Pearl River Basin (PeaRB).

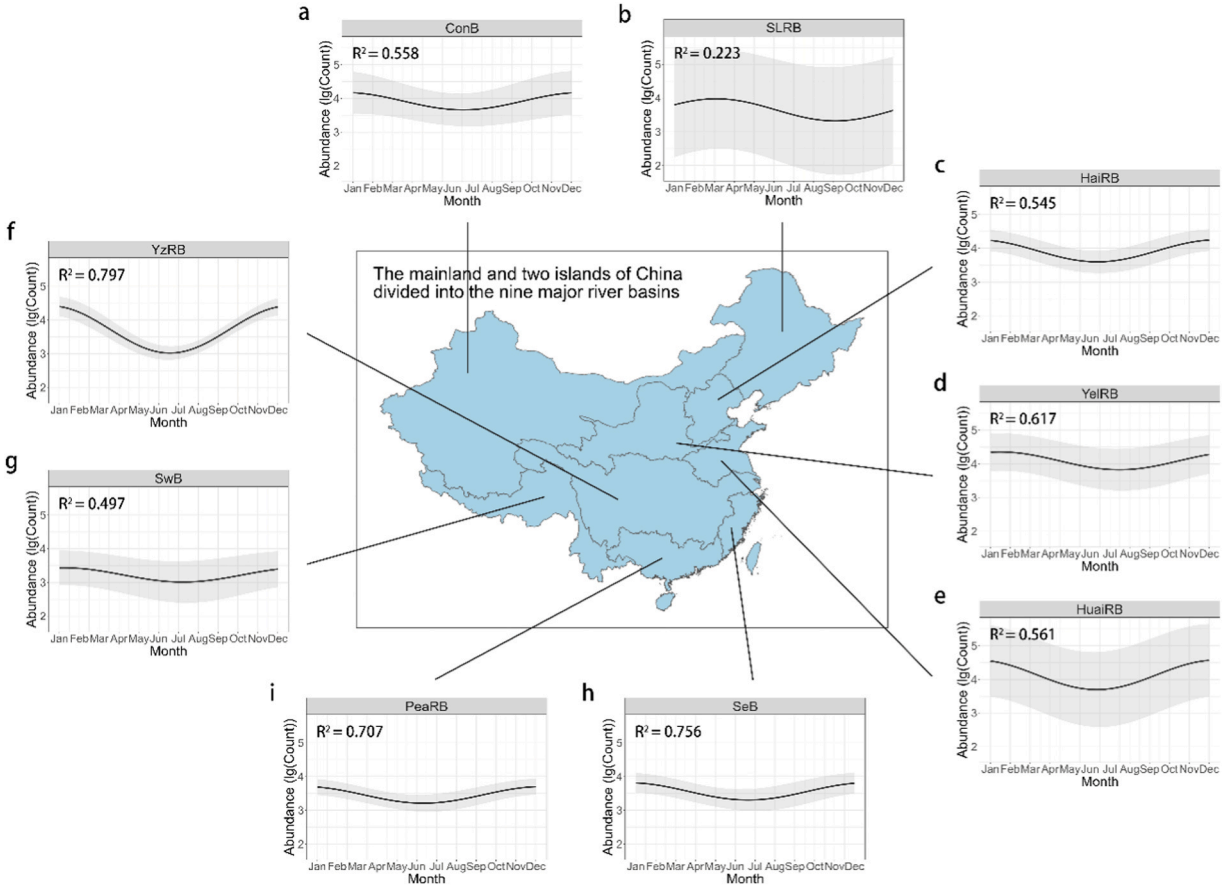



**Fig. 4.** Intra-annual variations of abundance in the waterbird community for each basin. a: the abundance variation of the Continental Basin (ConB); b: the abundance variation of the Songhua and Liaohe River Basin (SLRB); c: the abundance variation of the Haihe River Basin (HaiRB); d: the abundance variation of the Yellow River Basin (YelRB); e: the abundance variation of the Huaihe River Basin (HuaiRB); f: the abundance variation of the Yangtze River Basin (YzRB); g: the abundance variation of the Southwest Basin (SwB); h: the abundance variation of the Southeast Basin (SeB); i: the abundance variation of the Pearl River Basin (PeaRB).

The residential types that had high effects on the intra-annual abundance variation of the waterbird community in each basin were similar to those affecting species richness (Fig. 5B). Passage migrants or summer visitors had higher abundance and more similar variation curves to that of waterbird communities in ConB, SLRB, HaiRB, YelRB, and HuaiRB. Winter visitors had relatively greater effects on the intra-annual abundance variation of communities in the YzRB, SeB and PeaRB. In the SwB, there was no significant difference in the variation curve of each residential type.

The results of Simper analysis were consistent with the presentation of the variation curves in different residential types (Table A.4). The passage migrants and summer visitors were the most frequently displayed as the most important residential type for the community differences between two adjacent months in the ConB, SLRB, HaiRB, YelRB, and HuaiRB, while the winter visitors displayed more in the SeB and PeaRB. In the YzRB and SwB, no residential type had been identified as the most contributing assemblage significantly more often.

According to the result of the effect of different waterbird orders on the intra-annual variations of communities (Fig. 6), the order that affected community richness the most was different from that affected abundance the most even in the same basin. The Charadriiformes had the highest values of species richness index and the most similar richness variation curves to that of waterbird communities in all basins, while the Anseriformes had a greater effect on abundance variations of communities in the most basins (excepting the SwB, SeB, and PeaRB). The intra-annual richness trend of each order showed that seasonal variations of the Charadriiformes and Anseriformes in China were different. The richness of the Charadriiformes was higher in spring and autumn, while that of the Anseriformes was higher in winter.

According to the results of Simper analysis (Table A.4), the Charadriiformes and Anseriformes were identified as the order with the greatest contribution to the community differences between two adjacent months in most basins. The influence of Charadriiformes on richness was particularly strong, while the Anseriformes were more frequently identified as the most important order in abundance comparisons in the ConB, YelRB, HuaiRB and YzRB.

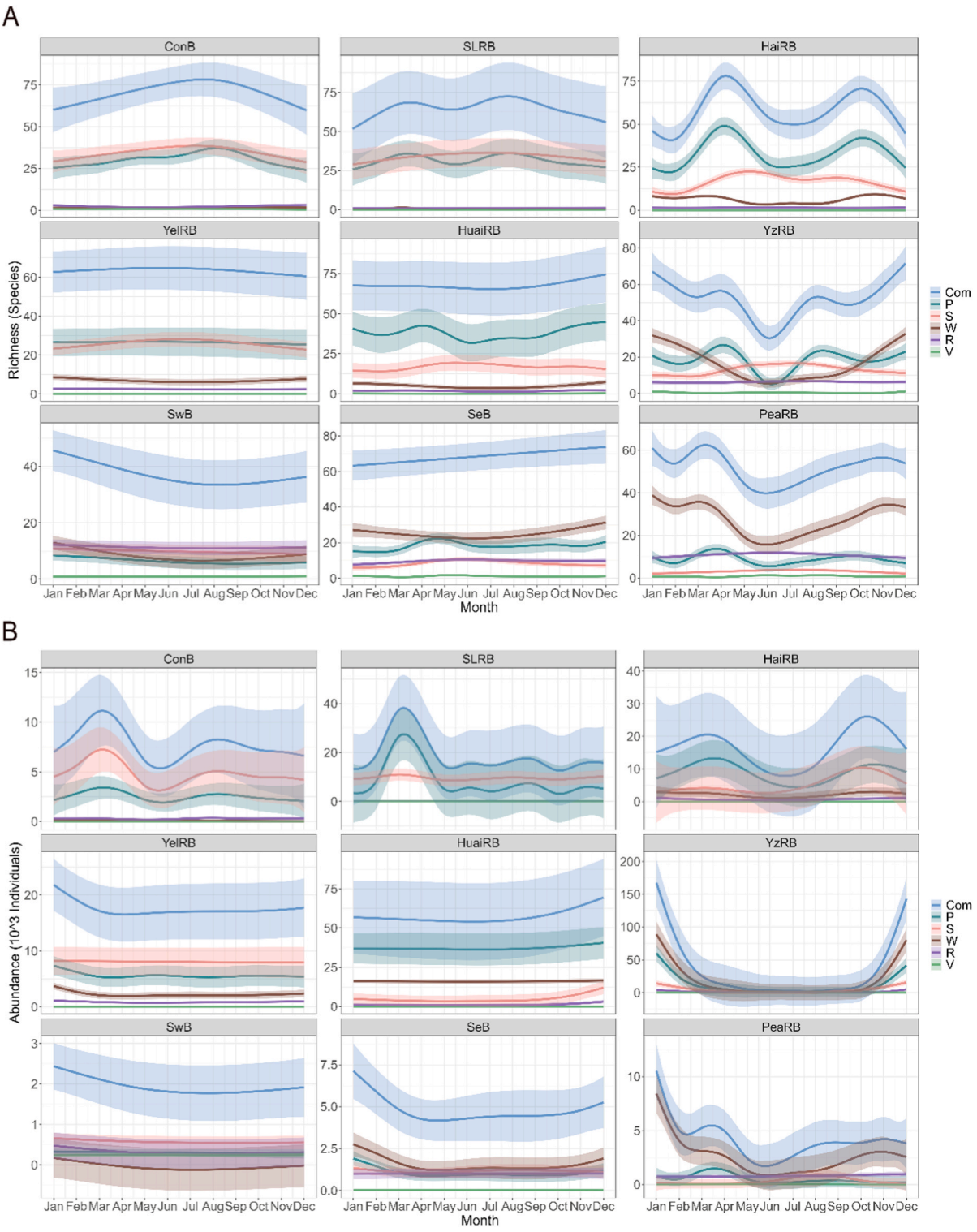



**Fig. 5.** Intra-annual variations of species richness (A) and abundance (B) among different residential types for each basin. The variation curve in blue was the intra-annual variation of the waterbird community in each basin (Com). Residential types include: passage migrant (P), summer visitor (S), winter visitor (W), resident (R), and vagrant visitor (V).

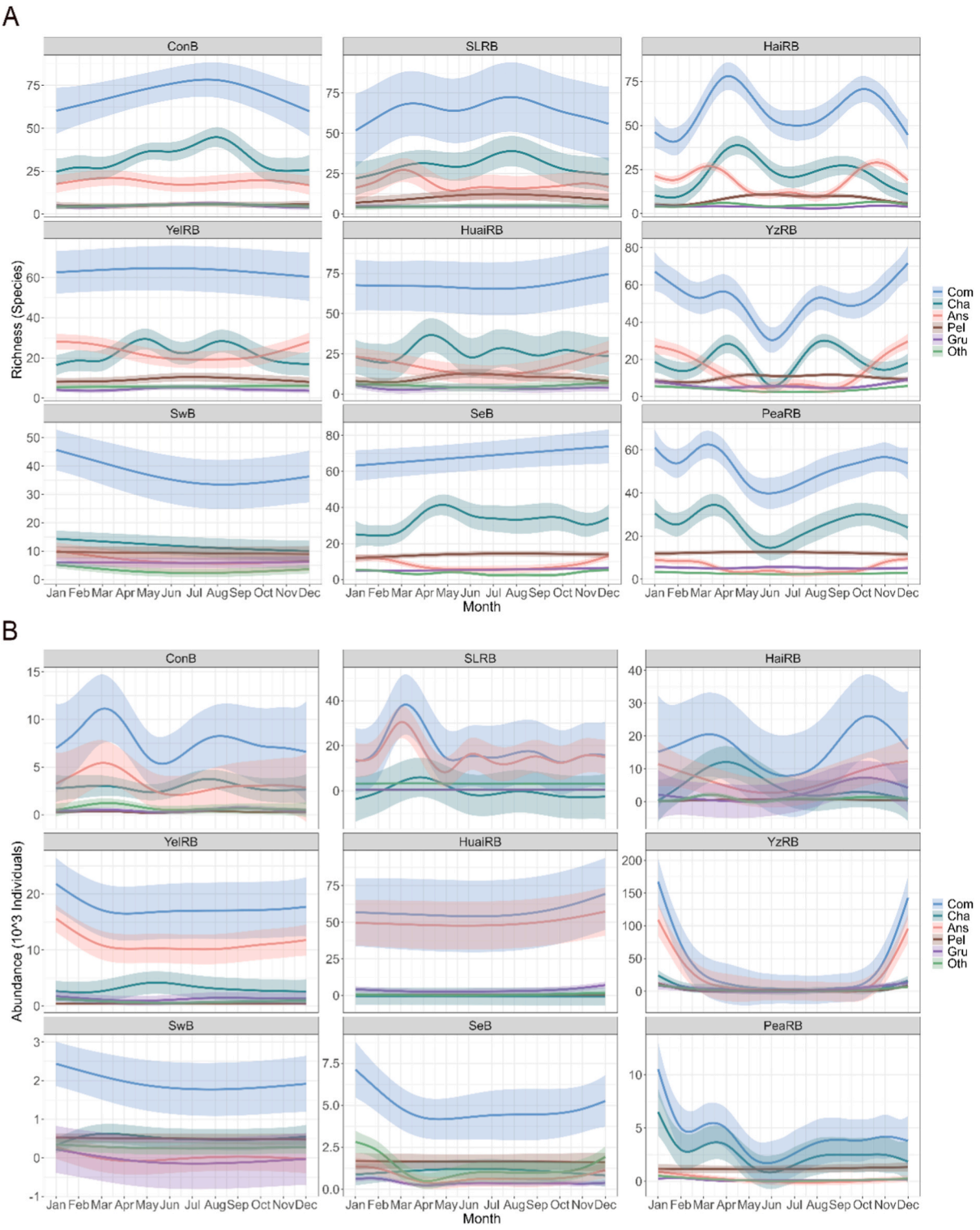



**Fig. 6.** Intra-annual variations of species richness (A) and abundance (B) among different waterbird orders for each basin. The variation curve in blue was the intra-annual variation of the waterbird community in each basin (Com). Orders include: Charadriiformes (Cha), Anseriformes (Ans), Pelecaniformes (Pel), Gruiformes (Gru), and Others (Oth).

## 4. Discussion

### 4.1. Seasonal variations of species compositions in waterbird communities

Our study results of seasonal waterbird communities among nine major river basins in China exhibited a higher similarity between the spring and autumn assemblages, while there were greater differences between the summer and winter assemblages. This was predominantly driven by species turnover due to waterbird migration. Seasonality is a key driver of bird migration (Dalby et al., 2014), birds move to areas with harsh winters to exploit a surplus in resources and avoid competition during the breeding season, and are driven by the avoidance of harsh winters during the non-breeding season (Somveille et al., 2015). Previous studies have indicated that only a few examples of species or populations have entirely distinct flyways in spring and autumn (Briedis et al., 2018). Therefore, in the same basin, the breeding and overwintering populations can vary greatly, while the spring and autumn assemblages consist of similar migratory species.

Interestingly, our results showed that the waterbird composition in the SwB remains stable throughout the year. This may be attributed to the region's limited wetland area and markedly different topography, which reduce the number of migratory routes for transiting waterbirds. On the contrary, the seasonal composition changes of the waterbird community in the YzRB were more prominent than in other basins. This is likely due to the YzRB has an important role in wintering ground for migratory waterbirds along the East Asia-Australasia Migratory Flyway, its variable wetland types that provide abundant food resources for numerous waterbirds (Chen et al., 2017; Gao et al., 2021). Wetlands in eastern China form part of the core areas of the East Asia-Australasia Migratory Flyway, and there have been many focused studies on the middle and lower YzRB (Lei et al., 2019; Yu et al., 2023). Our results of waterbird compositions among the nine basins showed the role of intra-annual community analysis in helping to reveal the seasonal regularity of waterbird migration, and also showed that not only the YzRB, most of the basins in China also have significant seasonal variations of waterbird communities, thus they also play an important role in waterbird migration. The well-being of a migratory waterbird population depends on the quantity and quality of the breeding areas, and at the stopover sites during migration and in the wintering range (Van Roomen et al., 2015). It follows that the large-scale and intra-annual waterbird population surveys are useful for the assessment of waterbird habitat and the development of effective and full annual cycle conservation strategies for migratory waterbirds in China (Lei et al., 2019).

### 4.2. The differences of waterbird communities among the nine basins

Previous studies suggested that the community should be largely influenced by dominant species (Smith and Knapp, 2003), thus, the dominant assemblages could indicate the similarities between regional waterbird communities. Our result showed there were slight and positive correlations between the regional differences of wetland area, EEQI and composition of waterbird community. It indicated that the greater disparities in wetland area or ecological environment quality among different basins lead to more pronounced differences in regional waterbird community composition. This is because the physical environment has a strong influence on local community structure (Ricklefs, 1987), the higher similarity between waterbird communities in adjacent basins was because the ecological environment quality of the basin was affected by adjacent basins (Ren and Lv, 2019). The waterbird communities in the SwB, SeB and PeaRB were particularly different from those in other basins, it might be because compared with other basins, these basins which are located in southern China had greater differences in physical characteristics, such as the obviously higher precipitation and river runoff (Zhang et al., 2011).

Wetlands are crucial for supporting the global migration of waterbirds as the wetland area is closely linked to resource abundance and distribution (Donnelly et al., 2020). In China, the enormous topographical variations and the outflow of rivers in the southern humid regions hinder wetland development (J et al., 2015), thus, although water is much more abundant along the southeast coast than in the northwest inland region (Chen et al., 2021), the wetland area is larger in the northern China. From our results of the trends of regional waterbird richness and abundance with regional wetland areas and EEQI, the basins with significantly larger wetland areas or better ecological environment quality had higher richness and abundance, which was consistent with the clear positive relationship between wetland size, habitat quality and waterbird species richness and abundance (Tavares et al., 2015). However, the YelRB had small wetland areas and low ecological environment quality but the high richness and abundance of the waterbird community, while the SeB and PeaRB had relatively high ecological environment quality but only the richness in SeB was high. This indicated that, in addition to wetland areas and ecological environment quality, many other factors might cause the species richness and abundance of communities in the SwB, SeB and PeaRB located in southern China to be lower than in other basins located in northern or central China: 1. The northern and central basins provide a suitable climate and abundant habitat resources, resulting in a substantial gathering of breeding waterbird species, while these waterbirds disperse to different southern regions, both within and outside the country, following various migratory routes southward. 2. Previous findings suggested that global-scale bird populations in the southern regions are declining, while those in the northern regions are benefiting from local climate warming (Amano et al., 2020). Thus, there may have been a northward shift in Chinese waterbird communities, leading to an increase in species richness and abundance of the northern and central basins (Jiguet et al., 2010; Li et al., 2021; Maclean et al., 2008). 3. Based on the species composition structures of nationwide waterbird communities, breeding assemblages were relatively concentrated in the northern and central basins, while wintering species were widely distributed in the southern China. The protected areas in China provide better coverage for breeding habitats than for wintering habitats (Cui et al., 2014; Prosser et al., 2018). The improved species and habitat conservation may contribute to the higher waterbird diversity in the northern and central basins. The impact of lost coastal wetlands and incomplete protection in wintering habitats on waterbirds is reflected in the lower overall values of species richness and

abundance in the southern basins. This highlights the need for better conservation planning of Chinese waterbirds and their habitats.

### 4.3. The effects of different assemblages on intra-annual variations of species richness and abundance in waterbird communities

The residential types explain the differences of intra-annual variation of species richness among basins. Basins in the cold-temperate and temperate zone have abundant breeding resources attracting summer visitors, while basins in the subtropical and tropical zone have suitable wintering area for various waterbird species (Prosser et al., 2018). Residents had a greater effect in the SwB, SeB and PeaRB that are located in south China because the stable primary productivity there allows certain waterbirds to breed, overwinter, and forage throughout the year, while migratory species prefer basins that have particularly harsh winters but less competition from resident species (Somveille et al., 2015). The population size of different species causes the inconsistent intra-annual variations of abundance with species richness in the waterbird community of the same basin. Our results showed that even within one basin, orders with the strongest effect on intra-annual variations of species richness and abundance were different. Charadriiformes play a leading role in affecting richness variation in almost all the nine basins, because it is the most species-rich order of waterbirds in China (Zheng, 2017; IUCN 2023; R Core Team 2023; Tallei et al. 2021; Trevor, Robert 1986; Wang et al. 2012; Wang et al. 2021; Xu et al. 2014). However, except for the SwB, SeB and PeaRB, Anseriformes lead abundance variations of waterbird communities with their larger population size. It is because the ConB and SLRB are located in northern China and serve as major breeding grounds for Chinese Anseriformes birds, and the HaiRB, YelRB, HuaiRB, and YzRB located in central China and provide rich habitats and plentiful food resources, including abundant lakes and rivers, for these swimming birds that live in open freshwater. According to the intra-annual variations of species richness of Charadriiformes and Anseriformes, the richness of Charadriiformes is higher in spring and autumn, while that of Anseriformes has higher richness in winter. Therefore, as Anseriformes affect the abundance variations of waterbird communities the most in most basins, the abundance in these basins is relatively high in winter.

Moreover, the detectability in different seasons or of different waterbird species also affects the community observation, especially the abundance. Many waterbird populations have different distribution behaviors during breeding or wintering seasons, and the assessment of population sizes and trends is better done in periods of the year when waterbirds congregate at relatively few sites (Van Roomen et al., 2015). For most species, the wintering ranges are smaller than breeding ranges (Newton, 2008). In winter, waterbirds aggregate in large numbers and can be observed more easily in accessible wetlands or coastal habitats. However, during the breeding season (in late spring or summer for most Chinese waterbirds), breeding species scatter throughout vast breeding areas in much lower densities, their concealed reproductive behaviors (Wang et al., 2012) and the sensitivity to disturbance make the detection much lower, which cause the low abundance of waterbird communities. In addition to the difficulty of observation, the congregation of breeding waterbirds preparing for migration (Warnock, 2010) and the entry of juveniles into the populations at the end of the breeding season also contributes to a relatively low abundance of waterbird communities in the middle of the year. What's more, although the threatened waterbird assemblage constituted a small proportion of the entire waterbird community, their intra-annual diversity variations were very similar with those of the community (Fig. A.4). It indicated that the threatened waterbird assemblage had similar regional and seasonal characteristics with that of the waterbird community analyzed earlier. Compared to other species, the threatened species are more sensitive to environmental changes, so they may rely more heavily on habitats and be more susceptible to human activities. Therefore, understanding their intra-annual variations is crucial for developing effective conservation policies of them. Different from the surveys of inter-annual variations of waterbird communities that focus on the effect of long-term environmental changes, the intra-annual variations show the role of seasonal migration and different populations in the parts of the waterbird life circle in shaping seasonal communities. Therefore, compared to the inter-annual variations, paying attention to intra-annual variations of waterbird communities can effectively improve our knowledge and conservation efforts for migratory waterbirds, and assess wetlands as breeding, overwintering or stopover sites. In addition, the differences of richness and abundance variations in the same basin suggested that multiple indices should be taken into consideration when calculating the variation of the waterbird community, in order to make a more comprehensive assessment of community status.

## 5. Conclusions

The seasonal migration of waterbird species created a seasonal cycle in the composition of the waterbird community. The intra-annual variations of species richness and abundance of waterbird communities varied across the nine major river basins of China and were primarily driven by different assemblages. The richness change of the waterbird community is directly influenced by the variation of the species composition, while the variability of community abundance is often more affected by the population sizes of different dominant assemblages and in different parts of the waterbird life circle (breeding, overwintering or stopover sites).

Climate warming has already caused changes in migration routes and a poleward movement of global waterbird communities. Serving as an important habitat for breeding and wintering, China plays an increasingly crucial role in the global avian migration. Our study on the intra-annual variations of the Chinese waterbird community highlighted the limitations of using a single diversity indicator to estimate community conditions, and also indicated that a specific assemblage can be useful in predicting community variations. The regional distribution and intra-annual variation of the Chinese waterbird community can aid in the better comprehension of waterbird migration patterns and guide the development of effective management and conservation strategies for global waterbird species and their habitats.

## Ethics statement

Not applicable: This manuscript does not include human or animal research.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data Availability

Data will be made available on request.

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## Appendix 1

**Table A.1**
Overview of Chinese nine major river basins. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

| River Basins | Area ($10^4$ km$^2$) | Catchment Area of Major Rivers ($10^4$ km$^2$) | Main Wetland Types | Wetland Area ($10^4$ km$^2$) | EEQI |
|---|---|---|---|---|---|
| ConB | ~ 333.89 | 12.99 | Saltwater Lake | 16.53 | 2.238 |
| SLRB | ~ 124.89 | 76.73 | Swamp | 11.20 | 2.847 |
| HaiRB | ~ 31.70 | 10.83 | Coast Wetland | 1.65 | 3.069 |
| YelRB | ~ 80.89 | 223.84 | Lake, River | 3.93 | 2.334 |
| HuaiRB | ~ 32.38 | 25.67 | Coast Wetland | 3.68 | 3.490 |
| YzRB | ~ 180.00 | 384.91 | Lake, River | 9.46 | 3.075 |
| SwB | ~ 85.26 | - | Swamp | 2.11 | 2.310 |
| SeB | ~ 24.02 | 7.27 | Coast Wetland | 1.86 | 3.883 |
| PeaRB | ~ 57.06 | 51.67 | Coast Wetland | 3.01 | 3.135 |

"-": Information about the catchment area of major rivers in southwest basin was incomplete.
EEQI: Ecological Environment Quality Index

**Table A.2**
The 253 waterbird species used in this study. Residential types: P=passage migrant; S=summer visitor; W=winter visitor; R=resident; V=vagrant visitor.

| Order | Common name | Species | IUCN | Residential types |
|---|---|---|---|---|
| Anseriformes | Lesser Whistling-duck | *Dendrocygna javanica* | LC | RSPV |
| Anseriformes | Brent Goose | *Branta bernicla* | LC | PWV |
| **Anseriformes** | **Red-breasted Goose** | ***Branta ruficollis*** | **VU** | **VW** |
| Anseriformes | Canada Goose | *Branta canadensis* | LC | V |
| Anseriformes | Barnacle Goose | *Branta leucopsis* | LC | V |
| Anseriformes | Cackling Goose | *Branta hutchinsii* | LC | V |
| Anseriformes | Bar-headed Goose | *Anser indicus* | LC | SWP |
| Anseriformes | Snow Goose | *Anser caerulescens* | LC | W |
| Anseriformes | Greylag Goose | *Anser anser* | LC | SWP |
| **Anseriformes** | **Swan Goose** | ***Anser cygnoid*** | **EN** | **SWPV** |
| Anseriformes | Bean Goose | *Anser fabalis* | LC | WP |
| Anseriformes | Thick-bill Bean Goose | *Anser serrirostris* | - | PW |
| Anseriformes | Greater White-fronted Goose | *Anser albifrons* | LC | WP |
| **Anseriformes** | **Lesser White-fronted Goose** | ***Anser erythropus*** | **VU** | **WPV** |
| Anseriformes | Mute Swan | *Cygnus olor* | LC | SPV |
| Anseriformes | Tundra Swan | *Cygnus columbianus* | LC | WP |
| Anseriformes | Whooper Swan | *Cygnus cygnus* | LC | SWPV |
| Anseriformes | Common Shelduck | *Tadorna tadorna* | LC | SWP |

(*continued on next page*)

**Table A.2** (*continued*)

| Order | Common name | Species | IUCN | Residential types |
|---|---|---|---|---|
| Anseriformes | Ruddy Shelduck | *Tadorna ferruginea* | LC | RSWP |
| Anseriformes | Mandarin Duck | *Aix galericulata* | LC | SWPV |
| Anseriformes | Cotton Pygmy-goose | *Nettapus coromandelianus* | LC | RSV |
| Anseriformes | Baikal Teal | *Sibirionetta formosa* | LC | WP |
| Anseriformes | Garganey | *Spatula querquedula* | LC | SWP |
| Anseriformes | Northern Shoveler | *Spatula clypeata* | LC | SWP |
| Anseriformes | Gadwall | *Mareca strepera* | LC | SWP |
| **Anseriformes** | **Falcated Duck** | ***Mareca falcata*** | **NT** | **SWP** |
| Anseriformes | Eurasian Wigeon | *Mareca penelope* | LC | SWP |
| Anseriformes | American Wigeon | *Mareca americana* | LC | V |
| **Anseriformes** | **Philippine Duck** | ***Anas luzonica*** | **VU** | **V** |
| Anseriformes | Indian Spot-billed Duck | *Anas poecilorhyncha* | LC | VR |
| Anseriformes | Chinese Spot-billed Duck | *Anas zonorhyncha* | LC | RSWPV |
| Anseriformes | Mallard | *Anas platyrhynchos* | LC | RSWP |
| Anseriformes | Northern Pintail | *Anas acuta* | LC | SWP |
| Anseriformes | Common Teal | *Anas crecca* | LC | SWP |
| Anseriformes | Red-crested Pochard | *Netta rufina* | LC | SWPV |
| **Anseriformes** | **Common Pochard** | ***Aythya ferina*** | **VU** | **SWP** |
| **Anseriformes** | **Baer's Pochard** | ***Aythya baeri*** | **CR** | **SWPV** |
| **Anseriformes** | **Ferruginous Duck** | ***Aythya nyroca*** | **NT** | **SWP** |
| Anseriformes | Tufted Duck | *Aythya fuligula* | LC | SWP |
| Anseriformes | Greater Scaup | *Aythya marila* | LC | WP |
| Anseriformes | Harlequin Duck | Histrionicus histrionicus | LC | P |
| **Anseriformes** | **Velvet Scoter** | **Melanitta fusca** | **VU** | **WPV** |
| **Anseriformes** | **Black Scoter** | **Melanitta americana** | **NT** | **V** |
| **Anseriformes** | **Long-tailed Duck** | ***Clangula hyemalis*** | **VU** | **WP** |
| Anseriformes | Common Goldeneye | *Bucephala clangula* | LC | SWP |
| Anseriformes | Smew | *Mergellus albellus* | LC | SWPV |
| Anseriformes | Goosander | *Mergus merganser* | LC | RSWPV |
| Anseriformes | Red-breasted Merganser | *Mergus serrator* | LC | SWPV |
| **Anseriformes** | **Scaly-sided Merganser** | ***Mergus squamatus*** | **EN** | **SWPV** |
| **Anseriformes** | **White-headed Duck** | ***Oxyura leucocephala*** | **EN** | **SW** |
| Gaviiformes | Red-throated Loon | *Gavia stellata* | LC | SWP |
| Gaviiformes | Arctic Loon | *Gavia arctica* | LC | WP |
| **Gaviiformes** | **Yellow-billed Loon** | ***Gavia adamsii*** | **NT** | **PW** |
| Procellariiformes | Wilson's Storm-petrel | *Oceanites oceanicus* | LC | V |
| **Procellariiformes** | **Short-tailed Albatross** | ***Phoebastria albatrus*** | **VU** | **SP** |
| **Procellariiformes** | **Swinhoe's Storm-petrel** | ***Hydrobates monorhis*** | **NT** | **S** |
| Procellariiformes | Tristram's Storm-petrel | *Hydrobates tristrami* | LC | V |
| Procellariiformes | Bonin Petrel | *Pterodroma hypoleuca* | LC | PV |
| **Procellariiformes** | **Streaked Shearwater** | ***Calonectris leucomelas*** | **NT** | **SPR** |
| Procellariiformes | Wedge-tailed Shearwater | *Ardenna pacificus* | LC | V |
| **Procellariiformes** | **Sooty Shearwater** | ***Ardenna grisea*** | **NT** | **SV** |
| Procellariiformes | Short-tailed Shearwater | *Ardenna tenuirostris* | LC | V |
| **Procellariiformes** | **Flesh-footed Shearwater** | ***Ardenna carneipes*** | **NT** | **SV** |
| Procellariiformes | Bulwer's Petrel | *Bulweria bulwerii* | LC | SP |
| Podicipediformes | Little Grebe | *Tachybaptus ruficollis* | LC | R |
| Podicipediformes | Red-necked Grebe | *Podiceps grisegena* | LC | SWP |
| Podicipediformes | Great Crested Grebe | *Podiceps cristatus* | LC | SWP |
| **Podicipediformes** | **Horned Grebe** | ***Podiceps auritus*** | **VU** | **SWP** |
| Podicipediformes | Black-necked Grebe | *Podiceps nigricollis* | LC | SWP |
| Phoenicopteriformes | Greater Flamingo | *Phoenicopterus roseus* | LC | P |
| Phaethontiformes | Red-billed Tropicbird | *Phaethon aethereus* | LC | S |
| Ciconiiformes | Painted Stork | *Mycteria leucocephala* | LC | S |
| Ciconiiformes | Asian Openbill | *Anastomus oscitans* | LC | V |
| Ciconiiformes | Black Stork | *Ciconia nigra* | LC | SW |
| **Ciconiiformes** | **Asian Woollyneck** | ***Ciconia episcopus*** | **NT** | **V** |
| Ciconiiformes | White Stork | *Ciconia ciconia* | LC | S |
| **Ciconiiformes** | **Oriental Stork** | ***Ciconia boyciana*** | **EN** | **SWP** |
| **Pelecaniformes** | **Black-headed Ibis** | ***Threskiornis melanocephalus*** | **NT** | **SWP** |
| **Pelecaniformes** | **Asian Crested Ibis** | ***Nipponia nippon*** | **EN** | **R** |
| Pelecaniformes | Glossy Ibis | *Plegadis falcinellus* | LC | P |
| Pelecaniformes | Eurasian Spoonbill | *Platalea leucorodia* | LC | SWP |
| **Pelecaniformes** | **Black-faced Spoonbill** | ***Platalea minor*** | **EN** | **SWP** |
| Pelecaniformes | Eurasian Bittern | *Botaurus stellaris* | LC | SWP |
| Pelecaniformes | Common Little Bittern | *Ixobrychus minutus* | LC | SP |
| Pelecaniformes | Yellow Bittern | *Ixobrychus sinensis* | LC | RSP |
| Pelecaniformes | Schrenck's Bittern | *Ixobrychus eurhythmus* | LC | SWP |
| Pelecaniformes | Cinnamon Bittern | *Ixobrychus cinnamomeus* | LC | RSP |
| Pelecaniformes | Black Bittern | *Ixobrychus flavicollis* | LC | RSPV |

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**Table A.2** (*continued*)

| Order | Common name | Species | IUCN | Residential types |
|---|---|---|---|---|
| **Pelecaniformes** | **White-eared Night-heron** | ***Gorsachius magnificus*** | **EN** | **R** |
| **Pelecaniformes** | **Japanese Night-heron** | ***Gorsachius goisagi*** | **VU** | **PW** |
| Pelecaniformes | Malay Night-heron | *Gorsachius melanolophus* | LC | R |
| Pelecaniformes | Black-crowned Night heron | *Nycticorax nycticorax* | LC | RS |
| Pelecaniformes | Rufous Night-heron | *Nycticorax caledonicus* | LC | V |
| Pelecaniformes | Green-backed Henron | *Butorides striata* | LC | RSWP |
| Pelecaniformes | Indian Pond-heron | *Ardeola grayii* | LC | V |
| Pelecaniformes | Chinese Pond-heron | *Ardeola bacchus* | LC | RSW |
| Pelecaniformes | Javan Pond-heron | *Ardeola speciosa* | LC | V |
| Pelecaniformes | Cattle Egret | *Bubulcus ibis* | LC | RSWP |
| Pelecaniformes | Gray Heron | *Ardea cinerea* | LC | RSWP |
| **Pelecaniformes** | **White-bellied Heron** | ***Ardea insignis*** | **CR** | **R** |
| Pelecaniformes | Purple Heron | *Ardea purpurea* | LC | RSWP |
| Pelecaniformes | Great White Egret | *Ardea alba* | LC | RSWP |
| Pelecaniformes | Intermediate Egret | *Ardea intermedia* | LC | RSWP |
| Pelecaniformes | Pied Heron | *Egretta picata* | LC | V |
| Pelecaniformes | Little Egret | *Egretta garzetta* | LC | RSW |
| Pelecaniformes | Pacific Reef-egret | *Egretta sacra* | LC | RP |
| **Pelecaniformes** | **Chinese Egret** | ***Egretta eulophotes*** | **VU** | **SWP** |
| Pelecaniformes | Great White Pelican | *Pelecanus onocrotalus* | LC | SWPV |
| **Pelecaniformes** | **Spot-billed Pelican** | ***Pelecanus philippensis*** | **NT** | **WP** |
| **Pelecaniformes** | **Dalmatian Pelican** | ***Pelecanus crispus*** | **NT** | **SWPV** |
| **Suliformes** | **Christmas Island Frigatebird** | ***Fregata andrewsi*** | **VU** | **P** |
| Suliformes | Lesser Frigatebird | *Fregata ariel* | LC | VSP |
| Suliformes | Masked Booby | *Sula dactylatra* | LC | S |
| Suliformes | Red-footed Booby | *Sula sula* | LC | RSWV |
| Suliformes | Brown Booby | *Sula leucogaster* | LC | RSWP |
| Suliformes | Little Cormorant | *Microcarbo niger* | LC | RS |
| Suliformes | Pelagic Cormorant | *Phalacrocorax pelagicus* | LC | WPV |
| Suliformes | Red-faced Cormorant | *Phalacrocorax urile* | LC | WV |
| Suliformes | Great Cormorant | *Phalacrocorax carbo* | LC | SWP |
| Suliformes | Japanese Cormorant | *Phalacrocorax capillatus* | LC | PW |
| **Gruiformes** | **Swinhoe's Rail** | ***Coturnicops exquisitus*** | **VU** | **SWP** |
| Gruiformes | Slaty-legged Crake | *Rallina eurizonoides* | LC | RSV |
| Gruiformes | Slaty-breasted Rail | *Lewinia striata* | LC | RS |
| Gruiformes | Eastern Water Rail | *Rallus indicus* | LC | RSWP |
| Gruiformes | Corncrake | *Crex crex* | LC | SPV |
| Gruiformes | Brown Crake | *Zapornia akool* | LC | R |
| Gruiformes | White-breasted Waterhen | *Amaurornis phoenicurus* | LC | RS |
| Gruiformes | Black-tailed Crake | *Zapornia bicolor* | LC | R |
| Gruiformes | Baillon's Crake | *Zapornia pusilla* | LC | SWP |
| Gruiformes | Spotted Crake | *Porzana porzana* | LC | SPV |
| Gruiformes | Ruddy-breasted Crake | *Zapornia fusca* | LC | RSP |
| **Gruiformes** | **Band-bellied Crake** | ***Zapornia paykullii*** | **NT** | **SPV** |
| Gruiformes | Watercock | *Gallicrex cinerea* | LC | S |
| Gruiformes | Purple Swamphen | *Porphyrio porphyrio* | LC | RV |
| Gruiformes | Common Moorhen | *Gallinula chloropus* | LC | RSP |
| Gruiformes | Common Coot | *Fulica atra* | LC | RSW |
| **Gruiformes** | **Siberian Crane** | ***Grus leucogeranus*** | **CR** | **SWP** |
| Gruiformes | Sandhill Crane | *Grus canadensis* | LC | W |
| **Gruiformes** | **White-naped Crane** | ***Grus vipio*** | **VU** | **SWPV** |
| Gruiformes | Demoiselle Crane | *Grus virgo* | LC | SWPV |
| **Gruiformes** | **Red-crowned Crane** | ***Grus japonensis*** | **VU** | **SWPV** |
| Gruiformes | Common Crane | *Grus grus* | LC | SWPV |
| **Gruiformes** | **Hooded Crane** | ***Grus monacha*** | **VU** | **SWPV** |
| **Gruiformes** | **Black-necked Crane** | ***Grus nigricollis*** | **NT** | **SWP** |
| **Charadriiformes** | **Great Thick-knee** | ***Esacus recurvirostris*** | **NT** | **W** |
| **Charadriiformes** | **Eurasian Oystercatcher** | ***Haematopus ostralegus*** | **NT** | **SWV** |
| Charadriiformes | Ibisbill | *Ibidorhyncha struthersii* | LC | R |
| Charadriiformes | Black-winged Stilt | *Himantopus himantopus* | LC | SWP |
| Charadriiformes | Pied Avocet | *Recurvirostra avosetta* | LC | SWP |
| **Charadriiformes** | **Northern Lapwing** | ***Vanellus vanellus*** | **NT** | **SWP** |
| **Charadriiformes** | **River Lapwing** | ***Vanellus duvaucelii*** | **NT** | **RW** |
| Charadriiformes | Grey-headed Lapwing | *Vanellus cinereus* | LC | SWP |
| Charadriiformes | Red-wattled Lapwing | *Vanellus indicus* | LC | R |
| Charadriiformes | Eurasian Golden Plover | *Pluvialis apricaria* | LC | V |
| Charadriiformes | Pacific Golden Plover | *Pluvialis fulva* | LC | WP |
| Charadriiformes | Grey Plover | *Pluvialis squatarola* | LC | WP |
| Charadriiformes | Common Ringed Plover | *Charadrius hiaticula* | LC | SWP |
| Charadriiformes | Long-billed Plover | *Charadrius placidus* | LC | WP |

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**Table A.2** (*continued*)

| Order | Common name | Species | IUCN | Residential types |
|---|---|---|---|---|
| Charadriiformes | Little Ringed Plover | *Charadrius dubius* | LC | RSWP |
| Charadriiformes | Kentish Plover | *Charadrius alexandrinus* | LC | RSWP |
| **Charadriiformes** | **Siberian Sandplover** | ***Charadrius mongolus*** | **EN** | **SWP** |
| Charadriiformes | Greater Sandplover | *Charadrius leschenaultii* | LC | SWP |
| Charadriiformes | Oriental Plover | *Charadrius veredus* | LC | SP |
| Charadriiformes | Eurasian Dotterel | *Eudromias morinellus* | LC | P |
| Charadriiformes | Greater Painted-snipe | *Rostratula benghalensis* | LC | RSW |
| Charadriiformes | Pheasant-tailed Jacana | *Hydrophasianus chirurgus* | LC | RS |
| Charadriiformes | Whimbrel | *Numenius phaeopus* | LC | WP |
| Charadriiformes | Little Curlew | *Numenius minutus* | LC | P |
| **Charadriiformes** | **Far Eastern Curlew** | ***Numenius madagascariensis*** | **EN** | **WP** |
| **Charadriiformes** | **Eurasian Curlew** | ***Numenius arquata*** | **NT** | **SWP** |
| **Charadriiformes** | **Bar-tailed Godwit** | ***Limosa lapponica*** | **NT** | **WP** |
| **Charadriiformes** | **Black-tailed Godwit** | ***Limosa limosa*** | **NT** | **SWP** |
| Charadriiformes | Ruddy Turnstone | *Arenaria interpres* | LC | WP |
| **Charadriiformes** | **Great Knot** | ***Calidris tenuirostris*** | **EN** | **WPV** |
| **Charadriiformes** | **Red Knot** | ***Calidris canutus*** | **NT** | **WP** |
| Charadriiformes | Ruff | *Calidris pugnax* | LC | WP |
| Charadriiformes | Broad-billed Sandpiper | *Calidris falcinellus* | LC | WP |
| **Charadriiformes** | **Sharp-tailed Sandpiper** | ***Calidris acuminata*** | **VU** | **WP** |
| **Charadriiformes** | **Curlew Sandpiper** | ***Calidris ferruginea*** | **NT** | **WP** |
| Charadriiformes | Temminck's Stint | *Calidris temminckii* | LC | WP |
| Charadriiformes | Long-toed Stint | *Calidris subminuta* | LC | WP |
| **Charadriiformes** | **Spoon-billed Sandpiper** | ***Calidris pygmeus*** | **CR** | **WP** |
| **Charadriiformes** | **Red-necked Stint** | ***Calidris ruficollis*** | **NT** | **WP** |
| Charadriiformes | Sanderling | *Calidris alba* | LC | WP |
| Charadriiformes | Dunlin | *Calidris alpina* | LC | WP |
| Charadriiformes | Little Stint | *Calidris minuta* | LC | P |
| Charadriiformes | White-rumped Sandpiper | *Calidris fuscicollis* | LC | V |
| Charadriiformes | Pectoral Sandpiper | *Calidris melanotos* | LC | P |
| **Charadriiformes** | **Asian Dowitcher** | ***Limnodromus semipalmatus*** | **NT** | **PV** |
| Charadriiformes | Long-billed Dowitcher | *Limnodromus scolopaceus* | LC | PWV |
| Charadriiformes | Eurasian Woodcock | *Scolopax rusticola* | LC | SWP |
| Charadriiformes | Jack Snipe | *Lymnocryptes minimus* | LC | WPV |
| Charadriiformes | Solitary Snipe | *Gallinago solitaria* | LC | SWP |
| **Charadriiformes** | **Wood Snipe** | ***Gallinago nemoricola*** | **VU** | **SV** |
| Charadriiformes | Pin-tailed Snipe | *Gallinago stenura* | LC | WP |
| Charadriiformes | Swinhoe's Snipe | *Gallinago megala* | LC | WP |
| Charadriiformes | Common Snipe | *Gallinago gallinago* | LC | SWP |
| Charadriiformes | Terek Sandpiper | *Xenus cinereus* | LC | WP |
| Charadriiformes | Red-necked Phalarope | *Phalaropus lobatus* | LC | WP |
| Charadriiformes | Red Phalarope | *Phalaropus fulicarius* | LC | WPV |
| Charadriiformes | Common Sandpiper | *Actitis hypoleucos* | LC | SWP |
| Charadriiformes | Green Sandpiper | *Tringa ochropus* | LC | SWP |
| Charadriiformes | Wandering Tattler | *Tringa incana* | LC | WP |
| **Charadriiformes** | **Grey-tailed Tattler** | ***Tringa brevipes*** | **NT** | **WP** |
| Charadriiformes | Lesser Yellowlegs | *Tringa flavipes* | LC | V |
| Charadriiformes | Common Redshank | *Tringa totanus* | LC | SWP |
| Charadriiformes | Marsh Sandpiper | *Tringa stagnatilis* | LC | SP |
| Charadriiformes | Wood Sandpiper | *Tringa glareola* | LC | SWP |
| Charadriiformes | Spotted Redshank | *Tringa erythropus* | LC | SWP |
| Charadriiformes | Common Greenshank | *Tringa nebularia* | LC | WP |
| **Charadriiformes** | **Spotted Greenshank** | ***Tringa guttifer*** | **EN** | **P** |
| Charadriiformes | Collared Pratincole | *Glareola pratincola* | LC | SV |
| Charadriiformes | Oriental Pratincole | *Glareola maldivarum* | LC | SP |
| Charadriiformes | Little Pratincole | *Glareola lactea* | LC | S |
| **Charadriiformes** | **Black-legged Kittiwake** | ***Rissa tridactyla*** | **VU** | **WP** |
| Charadriiformes | Slender-billed Gull | *Chroicocephalus genei* | LC | PSWV |
| Charadriiformes | Brown-headed Gull | *Chroicocephalus brunnicephalus* | LC | SWP |
| Charadriiformes | Black-headed Gull | *Chroicocephalus ridibundus* | LC | SWP |
| **Charadriiformes** | **Saunders's Gull** | ***Saundersilarus saundersi*** | **VU** | **SWP** |
| Charadriiformes | Little Gull | *Hydrocoloeus minutus* | LC | SPV |
| Charadriiformes | Ross's Gull | *Rhodostethia rosea* | LC | V |
| **Charadriiformes** | **Relict Gull** | ***Ichthyaetus relictus*** | **VU** | **SWP** |
| Charadriiformes | Pallas's Gull | *Ichthyaetus ichthyaetus* | LC | SWPV |
| Charadriiformes | Black-tailed Gull | *Larus crassirostris* | LC | SWP |
| Charadriiformes | Mew Gull | *Larus canus* | LC | WP |
| Charadriiformes | Glaucous-winged Gull | *Larus glaucescens* | LC | WP |
| Charadriiformes | Glaucous Gull | *Larus hyperboreus* | LC | WPV |
| Charadriiformes | Lesser Black-backed Gull | *Larus fuscus* | LC | WP |

(*continued on next page*)

**Table A.2** (*continued*)

| Order | Common name | Species | IUCN | Residential types |
|---|---|---|---|---|
| Charadriiformes | Arctic Herring Gull | *Larus smithsonianus* | LC | SWP |
| Charadriiformes | Caspian Gull | *Larus cachinnans* | LC | SW |
| Charadriiformes | Slaty-backed Gull | *Larus schistisagus* | LC | WP |
| Charadriiformes | Common Gull-billed Tern | *Gelochelidon nilotica* | LC | RSP |
| Charadriiformes | Caspian Tern | *Hydroprogne caspia* | LC | SWPR |
| Charadriiformes | Greater Crested Tern | *Thalasseus bergii* | LC | RS |
| Charadriiformes | Lesser Crested Tern | *Thalasseus bengalensis* | LC | P |
| **Charadriiformes** | **Chinese Crested Tern** | ***Thalasseus bernsteini*** | **CR** | **SWP** |
| Charadriiformes | Sandwich Tern | *Thalasseus sandvicensis* | LC | V |
| Charadriiformes | Little Tern | *Sternula albifrons* | LC | RS |
| **Charadriiformes** | **Aleutian Tern** | ***Onychoprion aleuticus*** | **VU** | **WP** |
| Charadriiformes | Bridled Tern | *Onychoprion anaethetus* | LC | RS |
| Charadriiformes | Sooty Tern | *Onychoprion fuscatus* | LC | WPV |
| Charadriiformes | Roseate Tern | *Sterna dougallii* | LC | RS |
| Charadriiformes | Black-naped Tern | *Sterna sumatrana* | LC | RSWP |
| Charadriiformes | Common Tern | *Sterna hirundo* | LC | SWP |
| **Charadriiformes** | **Black-bellied Tern** | ***Sterna acuticauda*** | **EN** | **R** |
| Charadriiformes | Whiskered Tern | *Chlidonias hybrida* | LC | SWP |
| Charadriiformes | White-winged Tern | *Chlidonias leucopterus* | LC | SWPV |
| Charadriiformes | Black Tern | *Chlidonias niger* | LC | SPV |
| Charadriiformes | Pomarine Jaeger | *Stercorarius pomarinus* | LC | P |
| Charadriiformes | Arctic Jaeger | *Stercorarius parasiticus* | LC | P |
| Charadriiformes | Long-tailed Jaeger | *Stercorarius longicaudus* | LC | P |
| Charadriiformes | Ancient Murrelet | *Synthliboramphus antiquus* | LC | SWPV |
| **Charadriiformes** | **Long-billed Murrelet** | ***Brachyramphus perdix*** | **NT** | **PV** |

"-": The category of Anser serrirostris was not found in the International Union for Conservation of Nature's (IUCN) Red List of Threatened Species.

**Table A.3**
The dominant species with Mcnaughton dominance index values ($\geq$0.02) among the nine major river basins in China. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

| Species | ConB | SLRB | HaiRB | YelRB | HuaiRB | YzRB | SwB | SeB | PeaRB |
|---|---|---|---|---|---|---|---|---|---|
| *Dendrocygna javanica* | | | | | | | | | 0.088 |
| *Anser indicus* | 1.263 | | | 2.009 | | 0.306 | 0.341 | | |
| *Anser anser* | 2.783 | 0.044 | 0.655 | 0.099 | | 1.439 | | | |
| *Anser cygnoid* | 0.065 | 0.917 | 0.070 | | | 1.284 | | 0.038 | |
| *Anser fabalis* | 0.288 | 1.484 | 1.654 | 0.670 | 0.319 | 2.489 | | | |
| *Anser serrirostris* | | 0.053 | 0.144 | 0.324 | 0.093 | 2.772 | | | |
| *Anser albifrons* | | 0.749 | | | | 0.724 | | | |
| *Anser erythropus* | | 0.035 | | | | 0.065 | | | |
| *Cygnus columbianus* | | | 0.659 | 0.030 | | 1.270 | | | |
| *Cygnus cygnus* | 0.586 | | 0.118 | 0.676 | | | | | |
| *Tadorna tadorna* | 1.192 | 0.287 | 1.026 | 0.386 | 0.123 | 0.103 | | | |
| *Tadorna ferruginea* | 2.571 | 0.438 | 1.361 | 3.067 | 0.100 | 0.621 | 1.672 | | 0.045 |
| *Aix galericulata* | | | 0.372 | 0.044 | 0.022 | 0.142 | | 0.568 | |
| *Sibirionetta formosa* | | 0.260 | | | 0.345 | | | | |
| *Spatula querquedula* | | | | | | | | | 0.052 |
| *Spatula clypeata* | 0.328 | | 0.263 | 0.063 | 0.172 | 0.059 | | 0.156 | 0.662 |
| *Mareca strepera* | 0.656 | 0.151 | 1.541 | 0.535 | 0.430 | 0.918 | 0.221 | | |
| *Mareca falcata* | | 0.083 | 0.586 | 0.273 | 1.984 | 2.788 | | 0.021 | |
| *Mareca penelope* | 0.189 | 0.039 | 0.032 | 0.158 | 0.193 | 0.430 | 0.041 | 0.274 | 0.286 |
| *Anas poecilorhyncha* | | | | | | | 0.853 | | |
| *Anas zonorhyncha* | 0.029 | 0.491 | 2.943 | 3.643 | 4.414 | 2.313 | 0.097 | 2.244 | 0.116 |
| *Anas platyrhynchos* | 4.853 | 2.094 | 6.441 | 4.450 | 3.436 | 1.513 | 0.206 | 0.150 | 0.027 |
| *Anas acuta* | 1.057 | 0.095 | 0.043 | 0.666 | 0.227 | 0.094 | | 0.031 | 0.084 |
| *Anas crecca* | 0.255 | 0.059 | 0.996 | 2.121 | 1.016 | 1.290 | 0.362 | 0.665 | 0.324 |
| *Netta rufina* | 2.228 | | | 0.508 | | 0.201 | 0.022 | | |
| *Aythya ferina* | 1.425 | 0.488 | 0.812 | 0.565 | 1.318 | 0.962 | 0.037 | | |
| *Aythya baeri* | | | | | 0.081 | 0.023 | | | |
| *Aythya nyroca* | 0.156 | | 0.078 | 0.463 | 0.050 | 0.196 | 0.136 | | |
| *Aythya fuligula* | 0.791 | 0.108 | 0.029 | 0.293 | 0.231 | 0.372 | 0.057 | 0.564 | 0.293 |
| *Bucephala clangula* | 0.134 | 0.080 | 0.164 | 0.529 | | 0.054 | | | |
| *Mergellus albellus* | 0.023 | 0.027 | 0.200 | 0.087 | 0.098 | 0.032 | | | |
| *Mergus merganser* | 0.575 | 0.189 | 1.097 | 0.877 | 0.305 | 0.142 | 0.128 | | |
| *Oxyura leucocephala* | 0.039 | | | | | | | | |

(*continued on next page*)

**Table A.3** (*continued*)

| Species | ConB | SLRB | HaiRB | YelRB | HuaiRB | YzRB | SwB | SeB | PeaRB |
|---|---|---|---|---|---|---|---|---|---|
| *Tachybaptus ruficollis* | 0.048 | 0.162 | 1.304 | 1.628 | 1.153 | 1.184 | 1.259 | 1.332 | 1.367 |
| *Podiceps cristatus* | 0.670 | 0.133 | 0.603 | 0.864 | 0.602 | 0.541 | 0.313 | 0.112 | 0.035 |
| *Podiceps nigricollis* | 0.175 | | 0.024 | 0.058 | | | | 1.371 | |
| *Anastomus oscitans* | | | | | | | 0.376 | | 0.170 |
| *Ciconia nigra* | 0.023 | | 0.045 | 0.059 | | | | | |
| *Ciconia boyciana* | | 0.056 | 0.183 | | | 0.260 | | | |
| *Platalea leucorodia* | 0.078 | 0.039 | 0.120 | 0.224 | 0.034 | 1.679 | | | |
| *Platalea minor* | | | | | | | | 0.101 | 0.088 |
| *Ixobrychus sinensis* | | | | | | | | 0.043 | 0.120 |
| *Ixobrychus cinnamomeus* | | | | | | | 0.053 | | 0.074 |
| *Nycticorax nycticorax* | | | 0.275 | 0.191 | 0.471 | 0.790 | 0.269 | 4.824 | 3.046 |
| *Butorides striata* | | | | | | | 0.106 | | 0.083 |
| *Ardeola bacchus* | | | 0.047 | 0.121 | 0.187 | 0.273 | 0.975 | 1.369 | 3.733 |
| *Bubulcus ibis* | | | | 0.035 | 0.232 | 0.235 | 2.481 | 1.095 | 1.163 |
| *Ardea cinerea* | 0.449 | 0.239 | 1.748 | 1.013 | 0.278 | 0.855 | 0.170 | 1.644 | 1.591 |
| *Ardea purpurea* | | | 0.026 | | | | | | |
| *Ardea alba* | 0.808 | 0.034 | 0.216 | 0.517 | 0.220 | 0.170 | 0.112 | 1.473 | 1.088 |
| *Ardea intermedia* | | | | | 0.020 | 0.105 | 0.148 | 0.082 | 0.300 |
| *Egretta garzetta* | | 0.047 | 0.362 | 0.556 | 0.916 | 1.277 | 3.105 | 7.219 | 6.639 |
| *Microcarbo niger* | | | | | | | 0.209 | | |
| *Phalacrocorax carbo* | 4.084 | 0.624 | 1.373 | 3.546 | 1.718 | 1.822 | 1.074 | 5.378 | 1.085 |
| *Zapornia akool* | | | | | | 0.033 | | | |
| *Amaurornis phoenicurus* | | | | | | 0.025 | 0.473 | 0.256 | 0.496 |
| *Zapornia bicolor* | | | | | | | 0.027 | | |
| *Zapornia fusca* | | | | | | | 0.029 | | |
| *Porphyrio porphyrio* | | | | | | | 2.231 | 0.052 | |
| *Gallinula chloropus* | 0.162 | | 0.187 | 0.657 | 1.248 | 0.921 | 2.702 | 0.955 | 1.077 |
| *Fulica atra* | 0.916 | 0.319 | 9.884 | 9.137 | 6.160 | 5.861 | 8.384 | 1.235 | 0.384 |
| *Grus leucogeranus* | | | | | | 0.147 | | | |
| *Grus virgo* | 0.383 | | | | | | | | |
| *Grus grus* | 0.754 | 0.040 | 0.874 | 0.404 | 0.267 | 0.117 | | | |
| *Grus monacha* | | | | | | 0.021 | | | |
| *Grus nigricollis* | 0.024 | | | 0.054 | | 0.085 | | | |
| *Haematopus ostralegus* | | 0.024 | | | | | | | |
| *Ibidorhyncha struthersii* | | | | | | | 0.043 | | |
| *Himantopus himantopus* | 0.609 | 0.086 | 0.598 | 0.457 | 0.150 | 0.170 | | 0.338 | 0.615 |
| *Recurvirostra avosetta* | 0.588 | 0.025 | 1.532 | 0.214 | 0.052 | 1.865 | | 0.335 | 1.639 |
| *Vanellus vanellus* | 0.918 | 0.155 | 0.252 | 0.098 | 0.029 | 0.232 | | | |
| *Vanellus duvaucelii* | | | | | | | 0.335 | | |
| *Vanellus cinereus* | 0.063 | | | 0.217 | 0.036 | 0.174 | 0.147 | 0.072 | 0.235 |
| *Vanellus indicus* | | | | | | | 0.260 | | |
| *Pluvialis fulva* | | | 0.051 | | | | | 0.112 | 0.747 |
| *Pluvialis squatarola* | | 0.072 | 0.054 | | 0.133 | | | 0.160 | 0.105 |
| *Charadrius placidus* | | | | 0.023 | | | | | |
| *Charadrius dubius* | 0.178 | 0.049 | 0.078 | 0.174 | 0.049 | 0.104 | 0.263 | 0.288 | 0.657 |
| *Charadrius alexandrinus* | 0.365 | 0.772 | 0.250 | 0.051 | 0.178 | 0.148 | | 0.739 | 5.779 |
| *Charadrius mongolus* | | | | | | | | 0.031 | 0.897 |
| *Charadrius leschenaultii* | | | | | | | | 0.380 | 1.148 |
| *Rostratula benghalensis* | | | | | | | 0.025 | | 0.029 |
| *Hydrophasianus chirurgus* | | | | | | 0.031 | | | |
| *Numenius phaeopus* | | 0.023 | | | | | | 0.052 | 0.050 |
| *Numenius madagascariensis* | | 0.782 | | | | | | | |
| *Numenius arquata* | 0.055 | 0.107 | 0.155 | | 0.170 | | | 0.171 | 0.120 |
| *Limosa lapponica* | | 0.474 | | | 0.057 | | | | |
| *Limosa limosa* | 0.347 | 0.319 | 1.116 | 0.055 | | 0.173 | | 0.096 | 0.151 |
| *Arenaria interpres* | | | | | | | | 0.048 | 0.023 |
| *Calidris tenuirostris* | | 2.218 | 0.054 | | 0.083 | | | 0.047 | 0.053 |
| *Calidris canutus* | | | 0.194 | | | | | 0.024 | 0.047 |
| *Calidris falcinellus* | | | | | | | | | 0.060 |
| *Calidris acuminata* | | | | | | | | 0.120 | |
| *Calidris ferruginea* | | | 0.114 | | | | | 0.043 | 0.221 |
| *Calidris temminckii* | 0.046 | | | | | 0.020 | | | 0.091 |
| *Calidris subminuta* | | | | | | | | | 0.038 |
| *Calidris ruficollis* | | | 0.053 | | 0.082 | 0.021 | | 0.379 | 0.987 |
| *Calidris alba* | | | | | | | | 0.035 | 0.489 |
| *Calidris alpina* | | 0.369 | 0.062 | | 0.743 | 0.356 | | 1.872 | 1.117 |
| *Calidris minuta* | 0.043 | | | | | | | | |
| *Gallinago gallinago* | | | 0.033 | 0.044 | 0.054 | 0.109 | 0.094 | 0.084 | 0.207 |
| *Xenus cinereus* | | | | | | | | | 0.047 |
| *Actitis hypoleucos* | 0.064 | | | 0.082 | | 0.059 | 0.188 | 0.284 | 0.335 |

(*continued on next page*)

**Table A.3** (*continued*)

| Species | ConB | SLRB | HaiRB | YelRB | HuaiRB | YzRB | SwB | SeB | PeaRB |
|---|---|---|---|---|---|---|---|---|---|
| *Tringa ochropus* | 0.021 | | 0.042 | 0.173 | 0.029 | 0.057 | 0.078 | 0.080 | 0.055 |
| *Tringa brevipes* | | | | | | | | 0.037 | |
| *Tringa totanus* | 0.588 | | | 0.259 | | | | 0.041 | 0.579 |
| *Tringa stagnatilis* | | | 0.251 | | 0.032 | | | 0.153 | 0.276 |
| *Tringa glareola* | | | 0.039 | 0.035 | | 0.094 | | 0.155 | 0.717 |
| *Tringa erythropus* | 0.036 | | 0.150 | 0.092 | 0.091 | 0.857 | | 0.055 | 0.090 |
| *Tringa nebularia* | | | 0.022 | | 0.036 | 0.070 | | 0.333 | 0.711 |
| *Glareola maldivarum* | | | 0.033 | | | | | | 0.158 |
| *Glareola lactea* | | | | | | | 0.603 | | |
| *Chroicocephalus brunnicephalus* | 0.446 | | | 1.486 | | 0.024 | 0.489 | | |
| *Chroicocephalus ridibundus* | 2.558 | 1.148 | 4.128 | 0.554 | 0.818 | 3.137 | 1.268 | 1.177 | 3.630 |
| *Saundersilarus saundersi* | | 0.076 | | | 0.061 | | | 0.086 | |
| *Ichthyaetus relictus* | 0.052 | 0.047 | 1.001 | 0.330 | | | | | |
| *Ichthyaetus ichthyaetus* | 0.829 | | | 0.437 | | | 0.067 | | |
| *Larus crassirostris* | | 1.186 | 0.025 | | | | | 0.255 | |
| *Larus canus* | | 0.025 | | | | | | | |
| *Larus smithsonianus* | | 0.270 | | | 0.040 | | | | |
| *Larus cachinnans* | 1.215 | | | | | | | | |
| *Gelochelidon nilotica* | | | | | | | | | 0.027 |
| *Hydroprogne caspia* | 0.026 | | | | | | | 0.048 | 0.149 |
| *Thalasseus bergii* | | | | | | | | 0.186 | |
| *Sternula albifrons* | 0.047 | | | | | | | 0.199 | |
| *Onychoprion anaethetus* | | | | | | | | | 0.260 |
| *Sterna hirundo* | 0.347 | 0.032 | 0.123 | 0.497 | 0.034 | 0.036 | | | |
| *Chlidonias hybrida* | 0.040 | | 0.212 | 0.081 | 0.052 | 0.699 | | 0.087 | 0.138 |
| *Chlidonias leucopterus* | 0.069 | | 0.055 | | | | | | 0.046 |

**Table A.4**

The assemblage that contributed the most to the differences between the waterbird communities of two adjacent months among the nine major river basins in China. Residential types include: passage migrant (P), summer visitor (S), winter visitor (W), resident (R), and vagrant visitor (V). Waterbird orders include: Charadriiformes (Cha), Anseriformes (Ans), Pelecaniformes (Pel), Gruiformes (Gru), and Others (Oth).Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

| | | Richness | | | | Abundance | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | average | sd | p | | average | sd | p |
| ConB | Jan - Feb | S | 0.142 | 0.095 | 0.957 | S | 0.384 | 0.191 | 0.645 |
| | Feb - Mar | S | 0.214 | 0.115 | 0.360 | S | 0.478 | 0.214 | 0.034* |
| | Mar - Apr | S | 0.182 | 0.122 | 0.740 | S | 0.443 | 0.198 | 0.158 |
| | Apr - May | P | 0.160 | 0.111 | 0.715 | S | 0.377 | 0.175 | 0.676 |
| | May - Jun | P | 0.174 | 0.097 | 0.509 | S | 0.347 | 0.158 | 0.901 |
| | Jun - Jul | S | 0.152 | 0.123 | 0.959 | S | 0.359 | 0.161 | 0.828 |
| | Jul - Aug | S | 0.164 | 0.185 | 0.900 | S | 0.398 | 0.174 | 0.469 |
| | Aug - Sep | S | 0.241 | 0.123 | 0.127 | S | 0.429 | 0.170 | 0.238 |
| | Sep - Oct | S | 0.102 | 0.066 | 0.998 | S | 0.343 | 0.184 | 0.866 |
| | Oct - Nov | S | 0.188 | 0.105 | 0.644 | S | 0.382 | 0.179 | 0.606 |
| | Nov - Dec | S | 0.176 | 0.115 | 0.737 | S | 0.451 | 0.230 | 0.226 |
| | Dec - Jan | S | 0.169 | 0.116 | 0.806 | S | 0.529 | 0.230 | 0.026* |
| | Jan - Feb | Ans | 0.183 | 0.129 | 0.048* | Ans | 0.460 | 0.241 | 0.046* |
| | Feb - Mar | Ans | 0.192 | 0.124 | 0.027* | Ans | 0.522 | 0.236 | 0.006** |
| | Mar - Apr | Cha | 0.163 | 0.100 | 0.927 | Ans | 0.428 | 0.190 | 0.101 |
| | Apr - May | Cha | 0.202 | 0.130 | 0.612 | Ans | 0.293 | 0.169 | 0.783 |
| | May - Jun | Cha | 0.191 | 0.107 | 0.766 | Ans | 0.263 | 0.157 | 0.922 |
| | Jun - Jul | Cha | 0.159 | 0.114 | 0.964 | Cha | 0.351 | 0.185 | 0.002** |
| | Jul - Aug | Cha | 0.188 | 0.170 | 0.784 | Cha | 0.320 | 0.193 | 0.020* |
| | Aug - Sep | Cha | 0.269 | 0.129 | 0.040* | Cha | 0.273 | 0.189 | 0.092. |
| | Sep - Oct | Cha | 0.137 | 0.077 | 0.996 | Ans | 0.354 | 0.237 | 0.422 |
| | Oct - Nov | Ans | 0.146 | 0.077 | 0.260 | Ans | 0.435 | 0.204 | 0.139 |
| | Nov - Dec | Ans | 0.195 | 0.156 | 0.094. | Ans | 0.568 | 0.274 | 0.024* |
| | Dec - Jan | Ans | 0.198 | 0.137 | 0.068. | Ans | 0.596 | 0.300 | 0.006** |
| SLRB | Jan - Feb | S | 0.160 | 0.087 | 0.980 | S | 0.285 | 0.211 | 0.469 |
| | Feb - Mar | P | 0.333 | 0.065 | 0.001*** | P | 0.811 | 0.136 | 0.001*** |
| | Mar - Apr | P | 0.192 | 0.125 | 0.722 | P | 0.409 | 0.336 | 0.782 |
| | Apr - May | S | 0.233 | 0.140 | 0.694 | P | 0.552 | 0.355 | 0.145 |
| | May - Jun | S | 0.271 | 0.203 | 0.335 | P | 0.343 | 0.331 | 0.927 |
| | Jun - Jul | S | 0.232 | 0.162 | 0.708 | P | 0.414 | 0.277 | 0.734 |
| | Jul - Aug | P | 0.219 | 0.130 | 0.426 | P | 0.439 | 0.278 | 0.647 |

(*continued on next page*)

**Table A.4** (*continued*)

| | | Richness | | | | Abundance | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | average | sd | p | | average | sd | p |
| | Aug - Sep | S | 0.243 | 0.146 | 0.601 | P | 0.460 | 0.282 | 0.541 |
| | Sep - Oct | S | 0.173 | 0.129 | 0.934 | P | 0.475 | 0.277 | 0.478 |
| | Oct - Nov | S | 0.319 | 0.184 | 0.091. | P | 0.376 | 0.256 | 0.813 |
| | Nov - Dec | S | 0.317 | 0.203 | 0.106 | S | 0.493 | 0.262 | 0.004** |
| | Dec - Jan | S | 0.271 | 0.155 | 0.357 | S | 0.448 | 0.186 | 0.013* |
| | Jan - Feb | Ans | 0.209 | 0.157 | 0.152 | Ans | 0.412 | 0.285 | 0.193 |
| | Feb - Mar | Ans | 0.275 | 0.047 | 0.006** | Ans | 0.714 | 0.136 | 0.001*** |
| | Mar - Apr | Ans | 0.171 | 0.130 | 0.462 | Ans | 0.495 | 0.217 | 0.033* |
| | Apr - May | Cha | 0.166 | 0.096 | 0.914 | Cha | 0.580 | 0.352 | 0.020* |
| | May - Jun | Cha | 0.202 | 0.121 | 0.633 | Cha | 0.373 | 0.313 | 0.574 |
| | Jun - Jul | Cha | 0.229 | 0.131 | 0.310 | Cha | 0.386 | 0.258 | 0.501 |
| | Jul - Aug | Cha | 0.241 | 0.133 | 0.203 | Cha | 0.617 | 0.286 | 0.007** |
| | Aug - Sep | Cha | 0.289 | 0.155 | 0.034* | Cha | 0.612 | 0.333 | 0.024* |
| | Sep - Oct | Cha | 0.144 | 0.083 | 0.957 | Cha | 0.499 | 0.289 | 0.151 |
| | Oct - Nov | Cha | 0.239 | 0.087 | 0.257 | Cha | 0.353 | 0.280 | 0.606 |
| | Nov - Dec | Ans | 0.328 | 0.189 | 0.001*** | Ans | 0.607 | 0.276 | 0.001*** |
| | Dec - Jan | Ans | 0.222 | 0.173 | 0.104 | Ans | 0.494 | 0.291 | 0.048* |
| HaiRB | Jan - Feb | P | 0.105 | 0.061 | 0.994 | P | 0.236 | 0.176 | 0.947 |
| | Feb - Mar | P | 0.226 | 0.075 | 0.152 | P | 0.407 | 0.170 | 0.029* |
| | Mar - Apr | P | 0.102 | 0.055 | 0.995 | P | 0.217 | 0.134 | 0.973 |
| | Apr - May | P | 0.080 | 0.059 | 1.000 | P | 0.380 | 0.174 | 0.118 |
| | May - Jun | P | 0.273 | 0.118 | 0.017* | P | 0.358 | 0.252 | 0.205 |
| | Jun - Jul | P | 0.144 | 0.106 | 0.893 | S | 0.276 | 0.155 | 0.067. |
| | Jul - Aug | P | 0.197 | 0.140 | 0.405 | S | 0.304 | 0.199 | 0.027* |
| | Aug - Sep | P | 0.248 | 0.161 | 0.065. | S | 0.322 | 0.288 | 0.005** |
| | Sep - Oct | P | 0.138 | 0.114 | 0.924 | P | 0.290 | 0.198 | 0.706 |
| | Oct - Nov | P | 0.130 | 0.099 | 0.958 | P | 0.295 | 0.160 | 0.676 |
| | Nov - Dec | P | 0.193 | 0.090 | 0.441 | P | 0.306 | 0.187 | 0.598 |
| | Dec - Jan | P | 0.123 | 0.075 | 0.971 | P | 0.290 | 0.179 | 0.718 |
| | Jan - Feb | Ans | 0.078 | 0.054 | 0.958 | Ans | 0.352 | 0.235 | 0.192 |
| | Feb - Mar | Cha | 0.143 | 0.056 | 0.712 | Cha | 0.286 | 0.252 | 0.175 |
| | Mar - Apr | Cha | 0.130 | 0.048 | 0.851 | Cha | 0.206 | 0.176 | 0.548 |
| | Apr - May | Ans | 0.071 | 0.041 | 0.981 | Cha | 0.341 | 0.197 | 0.061. |
| | May - Jun | Cha | 0.248 | 0.116 | 0.005** | Cha | 0.432 | 0.270 | 0.005** |
| | Jun - Jul | Cha | 0.159 | 0.114 | 0.516 | Cha | 0.276 | 0.219 | 0.204 |
| | Jul - Aug | Cha | 0.203 | 0.147 | 0.075. | Cha | 0.249 | 0.237 | 0.327 |
| | Aug - Sep | Cha | 0.238 | 0.162 | 0.007** | Cha | 0.237 | 0.315 | 0.393 |
| | Sep - Oct | Cha | 0.143 | 0.093 | 0.729 | Ans | 0.238 | 0.157 | 0.834 |
| | Oct - Nov | Cha | 0.093 | 0.070 | 0.987 | Ans | 0.303 | 0.157 | 0.446 |
| | Nov - Dec | Ans | 0.121 | 0.060 | 0.415 | Ans | 0.372 | 0.234 | 0.136 |
| | Dec - Jan | Ans | 0.091 | 0.061 | 0.865 | Ans | 0.425 | 0.201 | 0.031* |
| YelRB | Jan - Feb | P | 0.090 | 0.078 | 0.866 | S | 0.172 | 0.109 | 0.995 |
| | Feb - Mar | P | 0.098 | 0.071 | 0.773 | S | 0.215 | 0.139 | 0.950 |
| | Mar - Apr | P | 0.086 | 0.078 | 0.889 | S | 0.233 | 0.125 | 0.869 |
| | Apr - May | P | 0.137 | 0.048 | 0.321 | S | 0.239 | 0.126 | 0.866 |
| | May - Jun | P | 0.192 | 0.087 | 0.009** | P | 0.313 | 0.130 | 0.002** |
| | Jun - Jul | S | 0.126 | 0.129 | 0.160 | S | 0.349 | 0.218 | 0.083. |
| | Jul - Aug | P | 0.124 | 0.108 | 0.466 | S | 0.350 | 0.223 | 0.080. |
| | Aug - Sep | P | 0.136 | 0.091 | 0.331 | S | 0.348 | 0.248 | 0.094. |
| | Sep - Oct | P | 0.192 | 0.131 | 0.019* | S | 0.346 | 0.161 | 0.101 |
| | Oct - Nov | P | 0.169 | 0.107 | 0.078. | S | 0.328 | 0.173 | 0.179 |
| | Nov - Dec | P | 0.092 | 0.061 | 0.871 | S | 0.241 | 0.154 | 0.821 |
| | Dec - Jan | P | 0.081 | 0.066 | 0.933 | P | 0.204 | 0.106 | 0.615 |
| | Jan - Feb | Ans | 0.083 | 0.068 | 0.850 | Ans | 0.313 | 0.198 | 0.581 |
| | Feb - Mar | Ans | 0.084 | 0.061 | 0.798 | Ans | 0.323 | 0.175 | 0.479 |
| | Mar - Apr | Ans | 0.104 | 0.081 | 0.579 | Ans | 0.253 | 0.168 | 0.859 |
| | Apr - May | Cha | 0.126 | 0.055 | 0.314 | Cha | 0.223 | 0.133 | 0.091. |
| | May - Jun | Cha | 0.181 | 0.082 | 0.010** | Cha | 0.376 | 0.172 | 0.001*** |
| | Jun - Jul | Cha | 0.114 | 0.089 | 0.472 | Cha | 0.343 | 0.284 | 0.001*** |
| | Jul - Aug | Cha | 0.131 | 0.108 | 0.252 | Gru | 0.224 | 0.225 | 0.009** |
| | Aug - Sep | Cha | 0.137 | 0.095 | 0.185 | Cha | 0.226 | 0.148 | 0.111 |
| | Sep - Oct | Cha | 0.202 | 0.176 | 0.004** | Ans | 0.317 | 0.253 | 0.504 |
| | Oct - Nov | Ans | 0.142 | 0.132 | 0.123 | Ans | 0.370 | 0.187 | 0.189 |
| | Nov - Dec | Ans | 0.097 | 0.058 | 0.656 | Ans | 0.304 | 0.161 | 0.619 |
| | Dec - Jan | Ans | 0.080 | 0.061 | 0.876 | Ans | 0.399 | 0.195 | 0.101 |
| HuaiRB | Jan - Feb | P | 0.249 | 0.176 | 0.592 | P | 0.296 | 0.198 | 0.511 |
| | Feb - Mar | P | 0.225 | 0.187 | 0.781 | P | 0.282 | 0.178 | 0.637 |
| | Mar - Apr | P | 0.311 | 0.170 | 0.076. | P | 0.441 | 0.270 | 0.015* |

(*continued on next page*)

**Table A.4** (*continued*)

| | | Richness | | | | Abundance | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | average | sd | p | | average | sd | p |
| | Apr - May | P | 0.247 | 0.152 | 0.588 | P | 0.420 | 0.268 | 0.020* |
| | May - Jun | P | 0.281 | 0.120 | 0.227 | S | 0.384 | 0.178 | 0.093. |
| | Jun - Jul | P | 0.189 | 0.130 | 0.964 | S | 0.263 | 0.200 | 0.800 |
| | Jul - Aug | P | 0.204 | 0.148 | 0.866 | S | 0.309 | 0.220 | 0.511 |
| | Aug - Sep | S | 0.286 | 0.134 | 0.004** | S | 0.472 | 0.214 | 0.006** |
| | Sep - Oct | P | 0.275 | 0.144 | 0.314 | P | 0.331 | 0.158 | 0.315 |
| | Oct - Nov | P | 0.240 | 0.155 | 0.677 | S | 0.368 | 0.138 | 0.168 |
| | Nov - Dec | P | 0.266 | 0.182 | 0.409 | S | 0.394 | 0.147 | 0.085. |
| | Dec - Jan | P | 0.226 | 0.178 | 0.783 | S | 0.338 | 0.190 | 0.330 |
| | Jan - Feb | Ans | 0.163 | 0.133 | 0.281 | Ans | 0.419 | 0.183 | 0.013* |
| | Feb - Mar | Ans | 0.180 | 0.159 | 0.130 | Ans | 0.331 | 0.215 | 0.131 |
| | Mar - Apr | Cha | 0.279 | 0.152 | 0.026* | Cha | 0.431 | 0.321 | 0.003** |
| | Apr - May | Cha | 0.229 | 0.162 | 0.161 | Cha | 0.440 | 0.273 | 0.001*** |
| | May - Jun | Cha | 0.288 | 0.132 | 0.010** | Cha | 0.300 | 0.236 | 0.083. |
| | Jun - Jul | Cha | 0.212 | 0.149 | 0.325 | Cha | 0.237 | 0.294 | 0.245 |
| | Jul - Aug | Cha | 0.233 | 0.158 | 0.220 | Cha | 0.307 | 0.242 | 0.090. |
| | Aug - Sep | Cha | 0.267 | 0.185 | 0.056. | Pel | 0.331 | 0.174 | 0.002** |
| | Sep - Oct | Cha | 0.203 | 0.176 | 0.435 | Ans | 0.204 | 0.213 | 0.717 |
| | Oct - Nov | Ans | 0.185 | 0.134 | 0.095. | Ans | 0.334 | 0.249 | 0.127 |
| | Nov - Dec | Ans | 0.203 | 0.147 | 0.037* | Ans | 0.413 | 0.236 | 0.022* |
| | Dec - Jan | Ans | 0.157 | 0.116 | 0.341 | Ans | 0.386 | 0.260 | 0.039* |
| YzRB | Jan - Feb | W | 0.038 | 0.027 | 1.000 | W | 0.273 | 0.153 | 0.520 |
| | Feb - Mar | W | 0.042 | 0.039 | 1.000 | W | 0.234 | 0.115 | 0.810 |
| | Mar - Apr | P | 0.088 | 0.039 | 0.597 | W | 0.212 | 0.094 | 0.916 |
| | Apr - May | P | 0.111 | 0.127 | 0.210 | P | 0.178 | 0.125 | 0.817 |
| | May - Jun | P | 0.177 | 0.062 | 0.001*** | P | 0.201 | 0.126 | 0.506 |
| | Jun - Jul | P | 0.075 | 0.057 | 0.788 | S | 0.129 | 0.130 | 0.161 |
| | Jul - Aug | P | 0.124 | 0.087 | 0.103 | P | 0.157 | 0.129 | 0.934 |
| | Aug - Sep | P | 0.084 | 0.062 | 0.632 | P | 0.189 | 0.123 | 0.654 |
| | Sep - Oct | P | 0.081 | 0.059 | 0.713 | S | 0.172 | 0.121 | 0.015* |
| | Oct - Nov | W | 0.083 | 0.055 | 0.889 | W | 0.333 | 0.143 | 0.085. |
| | Nov - Dec | W | 0.051 | 0.021 | 1.000 | W | 0.339 | 0.132 | 0.057. |
| | Dec - Jan | P | 0.022 | 0.015 | 1.000 | W | 0.178 | 0.107 | 0.965 |
| | Jan - Feb | Cha | 0.038 | 0.027 | 0.997 | Ans | 0.313 | 0.209 | 0.506 |
| | Feb - Mar | Ans | 0.035 | 0.037 | 1.000 | Ans | 0.328 | 0.140 | 0.425 |
| | Mar - Apr | Cha | 0.113 | 0.046 | 0.254 | Ans | 0.209 | 0.148 | 0.926 |
| | Apr - May | Cha | 0.117 | 0.139 | 0.234 | Cha | 0.184 | 0.137 | 0.118 |
| | May - Jun | Cha | 0.197 | 0.064 | 0.001*** | Cha | 0.190 | 0.124 | 0.076. |
| | Jun - Jul | Cha | 0.197 | 0.064 | 0.001*** | Pel | 0.199 | 0.176 | 0.001*** |
| | Jul - Aug | Cha | 0.157 | 0.110 | 0.016* | Cha | 0.193 | 0.155 | 0.101 |
| | Aug - Sep | Cha | 0.107 | 0.076 | 0.352 | Cha | 0.254 | 0.165 | 0.009** |
| | Sep - Oct | Cha | 0.094 | 0.072 | 0.500 | Cha | 0.189 | 0.179 | 0.090. |
| | Oct - Nov | Ans | 0.097 | 0.070 | 0.549 | Ans | 0.357 | 0.166 | 0.235 |
| | Nov - Dec | Cha | 0.044 | 0.036 | 0.996 | Ans | 0.411 | 0.160 | 0.055. |
| | Dec - Jan | Cha | 0.032 | 0.023 | 1.000 | Ans | 0.227 | 0.139 | 0.862 |
| SwB | Jan - Feb | W | 0.063 | 0.041 | 0.961 | W | 0.169 | 0.097 | 0.383 |
| | Feb - Mar | W | 0.115 | 0.071 | 0.193 | R | 0.267 | 0.086 | 0.056. |
| | Mar - Apr | W | 0.076 | 0.051 | 0.887 | P | 0.203 | 0.173 | 0.004** |
| | Apr - May | R | 0.079 | 0.058 | 0.797 | P | 0.182 | 0.150 | 0.012* |
| | May - Jun | R | 0.123 | 0.085 | 0.061. | R | 0.231 | 0.144 | 0.233 |
| | Jun - Jul | S | 0.123 | 0.066 | 0.098. | R | 0.220 | 0.183 | 0.383 |
| | Jul - Aug | S | 0.145 | 0.103 | 0.009** | S | 0.237 | 0.134 | 0.013* |
| | Aug - Sep | S | 0.154 | 0.110 | 0.027* | S | 0.239 | 0.145 | 0.017* |
| | Sep - Oct | S | 0.111 | 0.065 | 0.248 | R | 0.207 | 0.150 | 0.489 |
| | Oct - Nov | W | 0.064 | 0.046 | 0.978 | W | 0.192 | 0.176 | 0.208 |
| | Nov - Dec | R | 0.082 | 0.061 | 0.697 | R | 0.180 | 0.114 | 0.795 |
| | Dec - Jan | W | 0.148 | 0.073 | 0.006** | R | 0.240 | 0.109 | 0.180 |
| | Jan - Feb | Ans | 0.064 | 0.047 | 0.931 | Gru | 0.244 | 0.180 | 0.284 |
| | Feb - Mar | Cha | 0.117 | 0.079 | 0.611 | Gru | 0.260 | 0.137 | 0.199 |
| | Mar - Apr | Cha | 0.107 | 0.084 | 0.768 | Cha | 0.280 | 0.191 | 0.001*** |
| | Apr - May | Cha | 0.106 | 0.079 | 0.817 | Cha | 0.240 | 0.159 | 0.005** |
| | May - Jun | Pel | 0.138 | 0.089 | 0.015* | Pel | 0.222 | 0.226 | 0.052. |
| | Jun - Jul | Pel | 0.127 | 0.093 | 0.047* | Ans | 0.220 | 0.195 | 0.112 |
| | Jul - Aug | Pel | 0.130 | 0.095 | 0.032* | Ans | 0.242 | 0.193 | 0.056. |
| | Aug - Sep | Cha | 0.172 | 0.108 | 0.080. | Ans | 0.259 | 0.224 | 0.065. |
| | Sep - Oct | Cha | 0.147 | 0.086 | 0.225 | Ans | 0.255 | 0.234 | 0.089. |
| | Oct - Nov | Cha | 0.091 | 0.065 | 0.931 | Gru | 0.217 | 0.222 | 0.437 |
| | Nov - Dec | Pel | 0.083 | 0.076 | 0.619 | Gru | 0.263 | 0.203 | 0.178 |

(*continued on next page*)

**Table A.4** (*continued*)

| | | Richness | | | | Abundance | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | average | sd | p | | average | sd | p |
| | Dec - Jan | Cha | 0.152 | 0.071 | 0.135 | Gru | 0.320 | 0.200 | 0.041* |
| SeB | Jan - Feb | W | 0.121 | 0.080 | 0.919 | W | 0.236 | 0.143 | 0.338 |
| | Feb - Mar | W | 0.127 | 0.094 | 0.901 | W | 0.239 | 0.129 | 0.305 |
| | Mar - Apr | P | 0.118 | 0.081 | 0.493 | W | 0.176 | 0.103 | 0.924 |
| | Apr - May | W | 0.086 | 0.065 | 0.997 | W | 0.151 | 0.101 | 0.981 |
| | May - Jun | P | 0.170 | 0.082 | 0.009** | R | 0.178 | 0.122 | 0.045* |
| | Jun - Jul | P | 0.110 | 0.068 | 0.659 | S | 0.195 | 0.140 | 0.054. |
| | Jul - Aug | P | 0.148 | 0.085 | 0.087. | S | 0.275 | 0.203 | 0.004** |
| | Aug - Sep | W | 0.138 | 0.075 | 0.776 | S | 0.273 | 0.226 | 0.001*** |
| | Sep - Oct | W | 0.168 | 0.099 | 0.322 | R | 0.219 | 0.111 | 0.004** |
| | Oct - Nov | W | 0.167 | 0.111 | 0.349 | W | 0.176 | 0.108 | 0.917 |
| | Nov - Dec | W | 0.121 | 0.087 | 0.894 | W | 0.216 | 0.102 | 0.584 |
| | Dec - Jan | W | 0.119 | 0.059 | 0.919 | W | 0.218 | 0.167 | 0.528 |
| | Jan - Feb | Cha | 0.140 | 0.087 | 0.980 | Oth | 0.202 | 0.150 | 0.209 |
| | Feb - Mar | Cha | 0.148 | 0.105 | 0.969 | Oth | 0.189 | 0.144 | 0.284 |
| | Mar - Apr | Cha | 0.227 | 0.151 | 0.365 | Cha | 0.252 | 0.189 | 0.300 |
| | Apr - May | Cha | 0.148 | 0.126 | 0.971 | Cha | 0.289 | 0.193 | 0.144 |
| | May - Jun | Cha | 0.291 | 0.155 | 0.023* | Pel | 0.340 | 0.263 | 0.009** |
| | Jun - Jul | Cha | 0.212 | 0.137 | 0.517 | Cha | 0.461 | 0.322 | 0.001*** |
| | Jul - Aug | Cha | 0.302 | 0.175 | 0.018* | Cha | 0.460 | 0.377 | 0.001*** |
| | Aug - Sep | Cha | 0.226 | 0.154 | 0.372 | Pel | 0.327 | 0.223 | 0.014* |
| | Sep - Oct | Cha | 0.258 | 0.169 | 0.110 | Pel | 0.371 | 0.229 | 0.003** |
| | Oct - Nov | Cha | 0.196 | 0.133 | 0.677 | Pel | 0.234 | 0.142 | 0.264 |
| | Nov - Dec | Cha | 0.145 | 0.081 | 0.973 | Oth | 0.261 | 0.168 | 0.037* |
| | Dec - Jan | Cha | 0.133 | 0.058 | 0.987 | Oth | 0.156 | 0.132 | 0.499 |
| PeaRB | Jan - Feb | W | 0.113 | 0.085 | 0.989 | W | 0.371 | 0.196 | 0.468 |
| | Feb - Mar | W | 0.112 | 0.097 | 0.988 | W | 0.166 | 0.115 | 1.000 |
| | Mar - Apr | W | 0.107 | 0.093 | 0.996 | W | 0.258 | 0.164 | 0.962 |
| | Apr - May | W | 0.217 | 0.144 | 0.244 | W | 0.404 | 0.191 | 0.239 |
| | May - Jun | W | 0.176 | 0.106 | 0.688 | R | 0.240 | 0.167 | 0.001*** |
| | Jun - Jul | W | 0.103 | 0.081 | 0.996 | W | 0.155 | 0.118 | 1.000 |
| | Jul - Aug | W | 0.152 | 0.116 | 0.839 | S | 0.237 | 0.243 | 0.001*** |
| | Aug - Sep | W | 0.108 | 0.093 | 0.986 | S | 0.235 | 0.234 | 0.002** |
| | Sep - Oct | W | 0.163 | 0.118 | 0.765 | W | 0.293 | 0.231 | 0.876 |
| | Oct - Nov | W | 0.046 | 0.040 | 1.000 | W | 0.300 | 0.204 | 0.863 |
| | Nov - Dec | W | 0.067 | 0.045 | 1.000 | W | 0.274 | 0.166 | 0.937 |
| | Dec - Jan | W | 0.128 | 0.080 | 0.959 | W | 0.449 | 0.207 | 0.071. |
| | Jan - Feb | Cha | 0.116 | 0.085 | 0.985 | Cha | 0.317 | 0.175 | 0.650 |
| | Feb - Mar | Cha | 0.130 | 0.099 | 0.960 | Cha | 0.209 | 0.202 | 0.989 |
| | Mar - Apr | Cha | 0.145 | 0.127 | 0.922 | Cha | 0.361 | 0.187 | 0.359 |
| | Apr - May | Cha | 0.296 | 0.184 | 0.004** | Cha | 0.489 | 0.273 | 0.006** |
| | May - Jun | Cha | 0.194 | 0.139 | 0.471 | Pel | 0.330 | 0.212 | 0.001*** |
| | Jun - Jul | Cha | 0.118 | 0.081 | 0.988 | Pel | 0.240 | 0.219 | 0.031* |
| | Jul - Aug | Cha | 0.220 | 0.145 | 0.247 | Cha | 0.323 | 0.283 | 0.610 |
| | Aug - Sep | Cha | 0.167 | 0.142 | 0.757 | Cha | 0.345 | 0.234 | 0.493 |
| | Sep - Oct | Cha | 0.185 | 0.163 | 0.577 | Cha | 0.298 | 0.219 | 0.765 |
| | Oct - Nov | Cha | 0.081 | 0.054 | 1.000 | Cha | 0.293 | 0.212 | 0.808 |
| | Nov - Dec | Cha | 0.098 | 0.065 | 0.997 | Cha | 0.253 | 0.174 | 0.938 |
| | Dec - Jan | Cha | 0.154 | 0.095 | 0.848 | Cha | 0.373 | 0.183 | 0.282 |

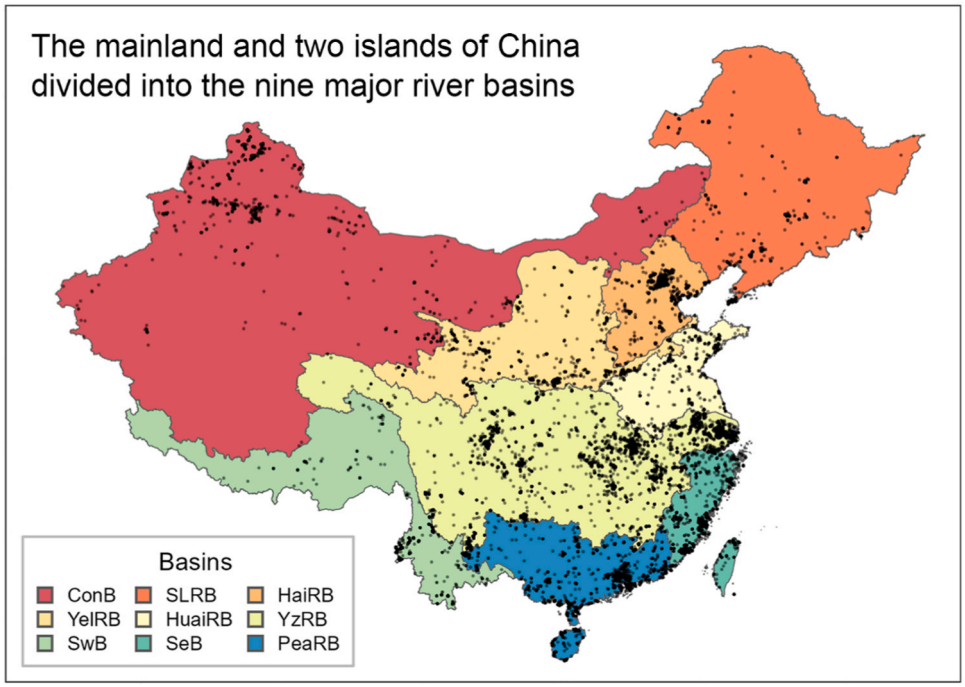



**Fig. A.1.** The observation points (representing as solid points) in nine major river basins of China. The study region included the mainland and two islands of China. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

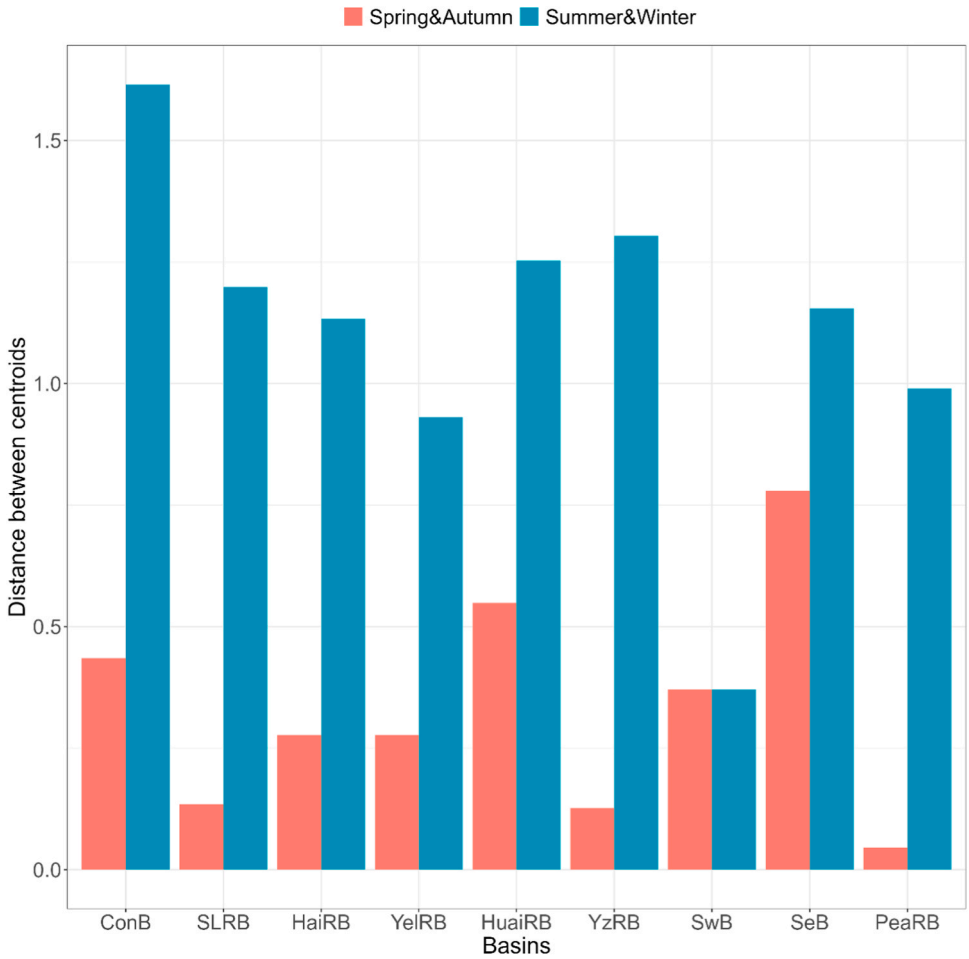



**Fig. A.2.** The distances between the centroids of spring&autumn waterbird communities and summer&winter waterbird communities among nine major river basins of China.

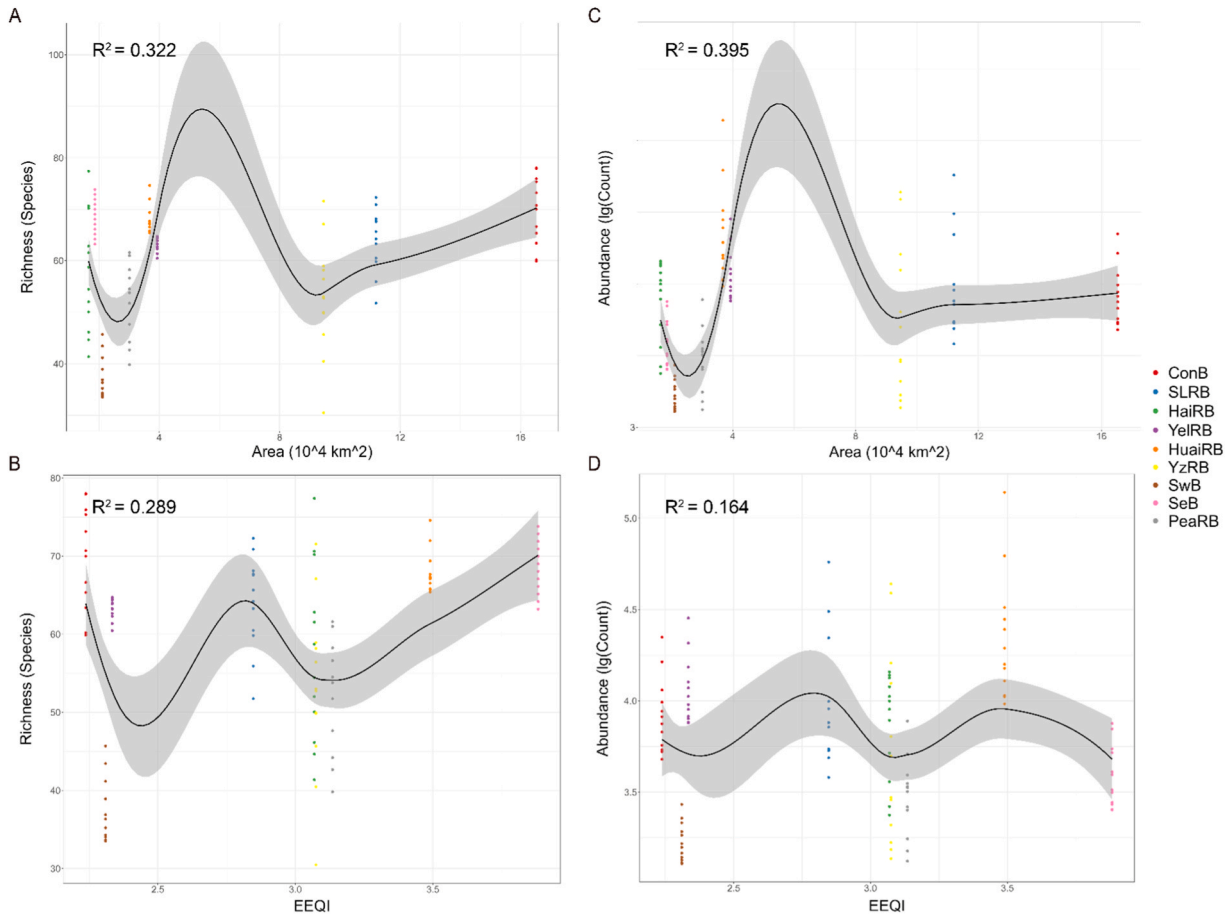



**Fig. A.3.** The trends of regional waterbird richness (A, B) and abundance (C, D) with regional wetland areas and Ecological Environment Quality Index (EEQI).

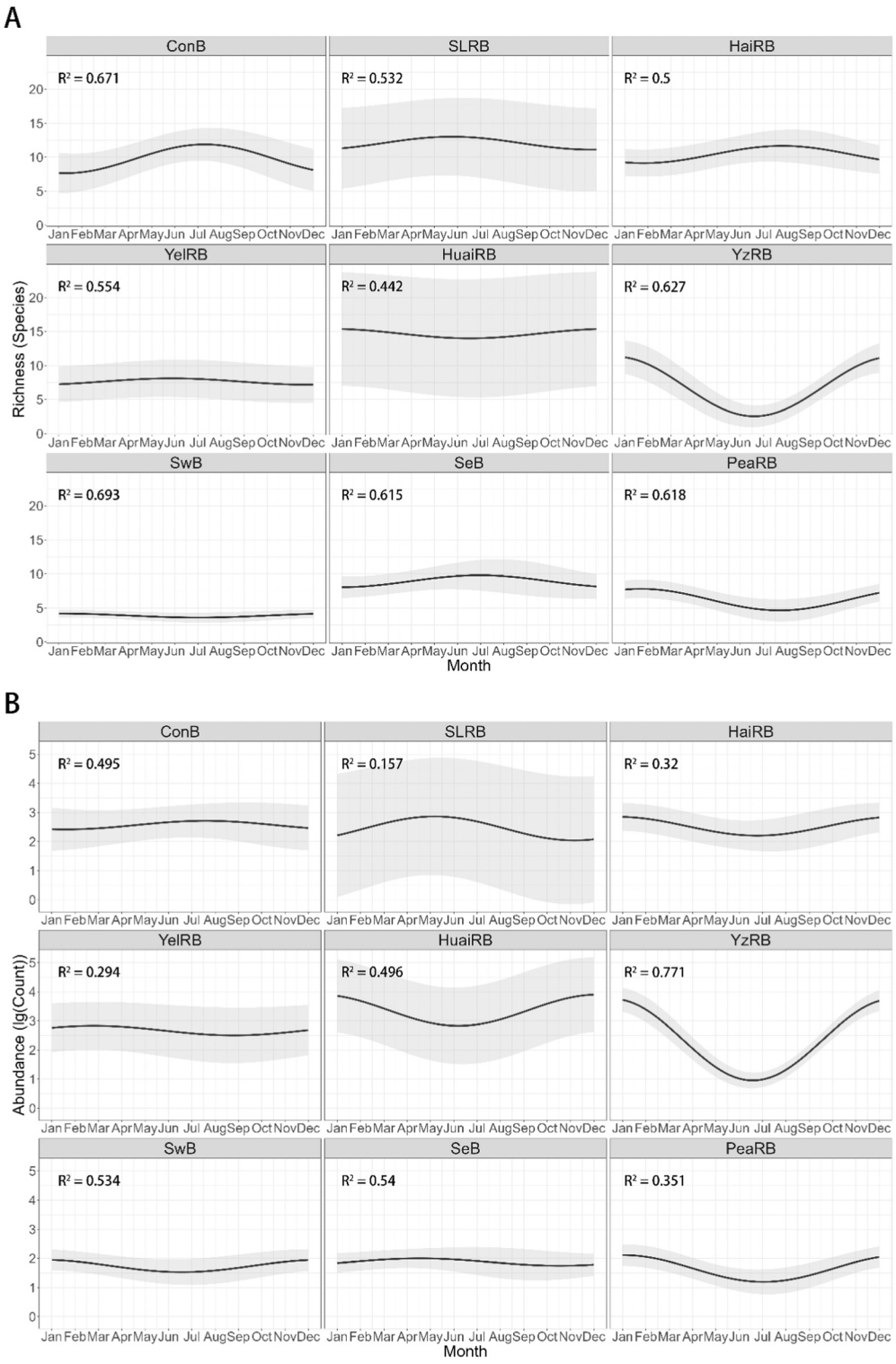



**Fig. A.4.** Intra-annual variations of species richness (A) and abundance (B) in the threatened waterbird assemblage for each basin. Basins include: the Continental Basin (ConB), the Songhua and Liaohe River Basin (SLRB), the Haihe River Basin (HaiRB), the Yellow River Basin (YelRB), the Huaihe River Basin (HuaiRB), the Yangtze River Basin (YzRB), the Southwest Basin (SwB), the Southeast Basin (SeB), and the Pearl River Basin (PeaRB).

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